

FROM SPACETIME FOAM TO BLACK-HOLE SEEDS AND BABY UNIVERSES: MERGER, SCATTERING, AND INFLATION OF TRAVERSABLE WORMHOLES

NIKITA M. SHIROKOV^{1,*} and ILYA NACHEVSKY^{2,†}

¹*Moscow Institute of Physics and Technology, Moscow, Russia*

²*Higher School of Economics, Moscow, Russia*

(Dated: September 26, 2026)

We evolve pairs of massive Ellis–Bronnikov drainholes—traversable wormholes supported by a phantom scalar field—in 3+1 numerical relativity, at equal masses and one throat compactness, without spin. An isolated throat is an unstable fixed point: the sign of its perturbation selects collapse or inflation, with an e-fold time within 2.5% of the González–Guzmán–Sarbach linear rate at our finest resolution. Like-signed scalar charges repel, so only opposite-signed pairs merge. The head-on merger forms a common horizon around both still-open throats, whose areal radius shrinks by a quarter as it swallows the phantom; the remnant rings down without a chirp. The spiral merger ends in a remnant of the same size, its common horizon recovered before a curvature wall that survives a $16\times$ refinement ladder and every gauge tried. No spinless, minimally coupled ghost-scalar wormhole can inspiral: outliving even the vacuum inspiral time from the innermost stable circular orbit needs an initial perturbation below 10^{-20} , so every encounter is a plunge. The loudest source is the horizonless fly-by, radiating $(4-7) \times 10^{-2} M$ in the dominant multipole, 16–29 times the vacuum merger control; the scalar field radiates comparable energy as a negative-energy $\ell = 1$ dipole that decays once a horizon forms. No throat of astrophysical mass survives, so these bursts are primordial and their detector is LISA, not LIGO: redshifted from birth at $z \approx 20$, they are millihertz bursts detectable one by one, every encounter above an inclination-averaged signal-to-noise ratio of 8 from 3×10^4 to $4 \times 10^6 M_\odot$ (optimal ratios 175–500 at 10^5 – $10^6 M_\odot$), while a lone collapse stays below. Binary-black-hole templates match the bursts (fitting factors 0.82–0.98), but their frequency falls or drifts where a Kerr remnant’s would hold, and a 127-template search of 2.26 h of LIGO O3b data finds no candidate. Collapse, alone or in a merger, converts a throat into a black hole of comparable mass, so a primordial population would supply halo-independent heavy seeds for the earliest supermassive black holes, evading the μ -distortion bound. The inflating half of the same population grows baby universes—anti-trapped regions advancing at $0.33 c$ behind exteriors of fixed mass, unable to inflate the parent universe—and their non-overlap today requires that growth to stall; the negative-energy scalar background the population leaves dilutes like radiation and cannot be the dark energy.

I. INTRODUCTION

Wheeler’s spacetime foam is a sea of handles—wormholes that open and close on the Planck scale [1–3]. Several mechanisms promote them to macroscopic size: inflation stretches a Planck-scale Lorentzian wormhole with the background [4]; bubble and domain-wall cosmologies produce post-inflationary regions joined through wormholes [5, 6]; and a gas of relic wormholes has been studied as a cosmology in its own right [7, 8]. None of this work evolves the aftermath: what a macroscopic relic wormhole does, and what two of them do to each other. That is the subject of this paper.

Two classical facts govern the dynamics. First, the simplest evolvable traversable wormhole—the Ellis–Bronnikov drainhole held open by a ghost scalar [9–12]—is unstable [14, 64]: depending on the sign of the perturbation, the throat collapses to a black hole or inflates, on a timescale set by the throat radius [13, 14], as 3+1 evolutions of the massless throat confirm non-

linearly [44]. Second, Geroch’s theorem forbids classical topology change [16, 17]: two wormholes cannot fuse into one, and none can be created classically. Creation is quantum [18], so our evolutions start at $t = 0$ from the newborn state—the stipulated initial data of every numerical binary, here with a physical origin attached.

Together these facts set selection rules. No astrophysical assembly channel exists: a throat decays faster than dynamical friction or radiation reaction could pair it. A primordial pair born from foam needs no assembly—birth and collision coincide, at a separation comparable to the objects—and its evolution is a race between orbital collision and throat decay, both rates fixed by the classical solution. The scalar adds a second rule: a phantom field repels like charges and attracts opposite ones, so like-signed pairs scatter while opposite-signed pairs fall together; and the sign of a lone throat’s perturbation selects collapse—a wormhole-to-primordial-black-hole channel beside the standard ones [19–21]—or inflation. A foam population thus splits into four fates—merger, scattering, collapse, inflation—across birth separation, perturbation and relative sign, the space mapped here. The binary evolved is the exterior approximation of a pair of handles (three asymptotic ends, the far sides passive), and what merges is not two wormholes into one

* shirokov.nm@phystech.edu

† ommnamashivaya@gmail.com

but two wormholes into one black hole containing both throats.

The observable is not a chirp but a population of bursts from the merger epoch, redshifted into whatever band the mass scale selects [22–25]. The epoch-independent deliverable is the energy radiated per event and its spectrum in throat-mass units (Sec. IX B). Every waveform behind it is extracted in the near zone, on spheres at $R = 10\text{--}44$, and none is extrapolated to null infinity: each radiated energy carries the spread across its spheres as a near-zone error bar (Sec. VIII). Collapse, alone or in a merger, also bears on JWST’s overmassive black holes at $z \gtrsim 4\text{--}6$, which favour heavy seeds of $10^4\text{--}10^6 M_\odot$ from formation channels that remain contested [46–49, 52]; Sec. X B develops it into a millihertz prediction, in which the encounter bursts—loudest for the fly-by and the spiral merger—are LISA sources and LIGO’s band stays empty.

Sections II and III give the model and numerics. Sections IV–VII follow the campaign from one throat to two: the lone wormhole, two throats at rest, and the head-on and spiral mergers. Section VIII treats the radiation of every channel, and Sec. IX confronts the rescaled waveforms with LIGO–Virgo–KAGRA data. Section X draws the consequences and states the scope; Sec. XI gives reproducibility and cost. Appendix A collects the code-health record—constraint evolution, the curvature wall under refinement and gauge, and device checks—and Appendix B supplementary measurements. All evolutions use GRTeclyn [41], the AMReX-based [42] GPU port of GRChombo [43], continuing Refs. [44, 45]. We use $G = c = 1$.

II. THE MODEL

A. Field equations

The matter is a single real, massless phantom scalar: gravity couples to *minus* the canonical stress tensor,

$$T_{ab} = -\left(\nabla_a \phi \nabla_b \phi - \frac{1}{2} g_{ab} \nabla_c \phi \nabla^c \phi\right), \quad V(\phi) \equiv 0, \quad (1)$$

so $\rho < 0$ where the gradient dominates—the exotic stress that holds a throat open. The field itself obeys the canonical wave equation $\square \phi = 0$, evolved in first-order form with $\Pi = (\partial_t \phi - \beta^i \partial_i \phi)/\alpha$.

B. Drainhole initial data

Each body is a massive Ellis–Bronnikov drainhole [9, 10]. In isotropic radius r about the throat centre, with

$$X = \frac{r - a^2/4r}{a}, \quad \Omega = 1 + \frac{a^2}{4r^2}, \quad u = \frac{m}{a} \left(\arctan X - \frac{\pi}{2} \right), \quad (2)$$

the exact static solution of Eq. (1) is

$$\alpha = e^u, \quad \gamma_{ij} = e^{-2u} \Omega^2 \delta_{ij}, \quad \phi = C \arctan X, \quad (3)$$

with $4\pi a^2 C^2 = a^2 + m^2$, $K_{ij} = 0$, $\Pi = 0$ and zero shift; in the evolved conformal variables $\chi = e^{2u} \Omega^{-2}$ and $h_{ij} = \delta_{ij}$. Both constraints hold identically at $t = 0$. Here a is the throat scale and m the ADM mass, carried by the lapse rather than the conformal factor, so χ at the throat stays in $[0.15, 0.25]$ for $m/a \leq 1$ and mass costs no resolution. The minimal surface sits at $r_t = \frac{1}{2}(m + \sqrt{m^2 + a^2})$, where $X = m/a$, so its areal radius is analytic: $R_{\min} = e^{-u(r_t)} \sqrt{a^2 + m^2} = \sqrt{a^2 + m^2} \exp[(m/a) \arctan(a/m)]$, which is a at $m = 0$. The production throat $a = 2$, $m = 1$ has $R_{\min} = \sqrt{5} e^{\frac{1}{2} \arctan 2} = 3.8895$ at $r_t = \frac{1}{2}(1 + \sqrt{5}) = 1.618$. The point $r \rightarrow 0$ is the far side’s spatial infinity, compactified ($\chi \sim r^4$). At $m = 0$ the throat is ultrastatic ($\alpha \equiv 1$, $M_{\text{ADM}} = 0$) and two such throats do not attract.

C. The binary superposition

Two throats at separation d (one-body radii r_A , r_B) are superposed in excess form,

$$u = u_A + u_B, \quad \psi = 1 + (\sqrt{\Omega_A} - 1) + (\sqrt{\Omega_B} - 1), \quad (4)$$

with $\chi = e^{2u} \psi^{-4}$, $\alpha = e^u$ and $\phi = \phi_A + \sigma \phi_B$, $\sigma = \pm 1$; the constant tail of the arctan profiles is subtracted so that $\phi \rightarrow 0$ at the boundary. Momentum enters through one Bowen–York term [35] per throat,

$$\hat{A}_{ij} = \frac{3}{2r^2} [P_i n_j + P_j n_i - (\delta_{ij} - n_i n_j) P \cdot n], \quad (5)$$

converted by $\tilde{A}_{ij} = \chi^{3/2} \hat{A}_{ij}$, with $K = 0$. The throats sit at $x = \mp d/2$ with $P^i = (0, \pm p, 0)$: momenta of magnitude p in units of the one-body mass, tangential to the separation and opposite on the two throats, so the pair carries no net momentum and orbital angular momentum $J = pd$; $p = 0$ is a release from rest. Since $\Pi = 0$ the matter momentum density vanishes and $\hat{A}_{ij}$ is flat-space divergence free, so the momentum constraint holds *exactly* at $t = 0$; only the Hamiltonian constraint carries the superposition defect, $O(a^2/d^2) + O(m/d)$. The relative sign σ is the pair’s one discrete parameter. Far from its throat each body is a mass m carrying the scalar charge $q = Ca$ ($\phi_A \simeq -q/r_A$ once the constant tail is subtracted), and a phantom mediator *repels* like charges and *attracts* opposite ones, so two throats at rest a large distance d apart attract with

$$F = \frac{m^2}{d^2} (1 - \sigma Q), \quad Q \equiv \left| \frac{F_\phi}{F_{\text{grav}}} \right| = \frac{4\pi q^2}{m^2} = \frac{a^2 + m^2}{m^2} > 1 \quad (6)$$

always ($Q = 5$ for the production throat): a like-signed pair repels and never merges under its own gravity, and $\sigma = -1$ is the merger channel. Two alternatives were

tested and not adopted: a windowed one-body correction of the Helfer type [36], which lowers the $t = 0$ defect but moves it into the fade zone between the throats, where it stalls the orbit instead of letting it plunge; and constraint-solved data from **GRtresa**, whose puncture decomposition cannot represent a throat. Every binary is therefore superposed, its defect declared rather than removed; what that defect radiates is bounded in Sec. VIII.

D. Declared seeds

The instability is probed with declared perturbations of the conformal factor only: a Gaussian shell on the minimal surface,

$$\psi \rightarrow \psi[1 + \varepsilon g(r)], \quad g = e^{-(r-r_t)^2/w^2}, \quad w = a/4, \quad (7)$$

which changes the throat's areal radius by a fraction $\approx 2\varepsilon$, and a quadrupolar factor

$$\psi \rightarrow \psi[1 + \varepsilon_2 g(r) P_2(\cos \theta)] \quad (8)$$

about the z axis through the throat. Neither seed is constraint-solved. K_{ij} and Π stay untouched, so the momentum constraint holds at $t = 0$; but the drainhole solves the linear equation $\nabla^2 \Psi = \pi |\nabla \phi|^2 \Psi$ ($\Psi^4 = \chi^{-1}$, flat operators), so the spherical seed leaves exactly $\mathcal{H} = -8\varepsilon \Psi^{-4}(1 + \varepsilon g)^{-5}(\nabla^2 g + 2\nabla \ln \Psi \cdot \nabla g)$ —linear in ε , confined to the shell and, the shell being narrow, not small there: its peak is $0.93 \times 16\pi|\rho|$ at $|\varepsilon| = 10^{-2}$. The quadrupolar seed's defect is pure $\ell = 2$ at first order. CCZ4 feeds the defect into Θ and, within the first steps, into $\mathcal{M}$ (Fig. 10(b)). Above $\varepsilon \simeq 0.025$ the spherical seed turns the minimal surface into a maximum between two minima. The spherical seed selects the branch but cannot make waves: a spherical collapse has no $\ell \geq 2$ moment and radiates no gravitational waves. The quadrupolar seed is there for that reason. It has no $\ell = 0$ part at first order, so it does not compete with the spherical seed for the branch, and it gives a collapsing throat a quadrupole to shed—the only gravitational-wave signal a lone throat has (Sec. VIII A).

III. NUMERICAL SETUP

The campaign is one-knob: every run differs from a named reference in a single setting, or repeats one with diagnostics, checkpoints or time added, and has one line—difference, outcome, caveats—in the registry shipped with the results (Sec. XI). Table I groups the 145 evolutions; 12 form an archived shakedown that fixed the gauge, dissipation and tagging choices below and are not used for physics, leaving 133.

A. Evolution scheme

We evolve CCZ4 [37] with damping $(\kappa_1, \kappa_2, \kappa_3) = (3, 0, 1)$, 1+log slicing [38] started from the exact static lapse of Eq. (3), a Gamma-driver shift [39, 40] with $\eta = 1$ started from zero, fourth-order stencils, RK4 and Kreiss–Oliger dissipation $\sigma = 0.1$. The time step is $\Delta t = 0.02 \Delta x$, an order of magnitude below the usual CCZ4 Courant factor, because larger steps fail: a multiplier of 0.05 departs from the production run at $t \approx 33$, and 0.1 dies at $t = 16$.

B. Grid, boundaries and regularisation

The domain is a cube of side $L = 64$ centred on the system, with a 128^3 base grid ($\Delta x = 0.5$). Nested boxes, halving per level and centred on the tracked throats (half-width 4 per throat at the production finest level), are redrawn every 16 steps. Production runs use three refinement levels (finest $\Delta x = 1/16$); resolution arms go to levels 4–7 where stated, and the production spiral, the level-5 fly-by and two lone-throat arms use $L = 128$: the same cells in a box twice as wide. Outer boundaries are Sommerfeld, backed by a spherical sponge about the box centre over the outer quarter of the half-width— $r = 24$ –32 at $L = 64$ and 48–64 at $L = 128$ —in which extra dissipation ramps to $\sigma_{\text{sp}} = 4$ as the fourth power of the depth; every extraction sphere lies inside it. The vacuum controls (Sec. VIII E) instead run **GRTEclyn**'s binary-black-hole example with its own numerical settings and no sponge.

Each throat carries a compactified second infinity at its centre, where $\chi \sim r^4$, so the evolved variables are floored at $\chi \geq 10^{-8}$ and $\alpha \geq 10^{-10}$ (a variant floors χ at 10^{-20} and applies 10^{-8} only where $1/\chi$ is evaluated). Two further devices act only inside regions already trapped: exponential damping of the scalar where the lapse has collapsed, and an interior freeze that multiplies every right-hand side by $1 - W(r)$, with W a C^2 quintic step between radii r_{full} and r_{start} . All three are numerical devices, each measured by a twin: χ -floor twins at 10^{-8} and 10^{-20} re-converge to four digits after touching the floor, the scalar damping moves the spiral's wall by less than a unit (Sec. VII C), and the freeze leaves the exterior unchanged (Secs. VI B and VI B).

C. Diagnostics

Every coarse step the code writes the volume-averaged norms $\|\mathcal{H}\| = (\int \mathcal{H}^2 dV/L^3)^{1/2}$ and $\|\mathcal{M}\| = (\int \mathcal{M}_i \mathcal{M}_i dV/L^3)^{1/2}$, evaluated with fourth-order differences on the base grid over the whole domain—sponge and throat interiors included—after the finer levels have been restricted onto it; they depend on the box and are compared only within one. Both throat centres are tracked, and about each the areal radius $R(r)$ is scanned on coordinate spheres; its minimum is the throat. The

TABLE I. The run matrix: the 145 packed evolutions in groups, each varying one knob against a named reference or repeating one; per-run lines are in the shipped registry. p is each throat’s tangential Bowen–York momentum (Sec. II C). Unless stated, $a = 2$, $m = 1$ ($R_{\min} = 3.89$), $\sigma = -1$, $d = 12$ and $L = 64$. Left: the single throat, the pair, the head-on and the controls. Right: the orbital family.

group	knob(s)	runs	Sec.	group	knob(s)	runs	Sec.
lone throat	levels 2–4; χ floor; Δt ; $L=128$	9	IV	momentum scan	$p = 0.20\text{--}0.45$	4	VII A
spherical kicks	$\varepsilon = \pm 10^{-3} \dots \pm 10^{-1}$; levels 3–5	10	IV C	fly-by, $p = 0.45$	level 3; level 5 on the doubled box	2	VIII D
quadrupolar kicks	$\varepsilon_2 = 0.005\text{--}0.05$; kick-free arm	6	VIII A	orbital chain, $p = 0.12$	N ; levels; damping; dissipation; χ floor	13	VII
scalar-stream re-runs	unkicked level 4, $L=64/128$; $\varepsilon=+10^{-2}$	3	VIII G	refinement ladder	levels 4–7; Δt ; damping	8	VII C
two throats at rest	σ ; $a = 1\text{--}3$; $d = 12\text{--}18$; $p = 0.12$ push	9	V	interior freeze, $L=64$	skin radii; arming time; levels; extraction	7	VIII C
placement probes	$d = 6\text{--}48$, one step each	18	VC	wall probes	$p = 0.15\text{--}0.25$; levels; fill off	6	VII C
head-on, $d = 8/12$	fill; levels 3/5; down-step; $\pm\varepsilon$	10	VI	production spiral, $L=128$	three-stage chain; fill-window twin; level 5 from $t=0$; mouth scans	6	VII B
half-mass, $m = 0.5$	pair at $d = 6$; lone control	2	VIC	Helper twins	correction; window; levels	6	IIC
vacuum BBH	same d , p , masses; no scalar, no sponge	3	VIII E	gauge arms	1+log halved; harmonic-class; $\eta=4$; spiral, head-on	10	VII C
constraint-solved data	GRtresa bridge	1	IIC	shakedown (archived)	gauge, dissipation, tagging; boosted throat	12	III A

spheres run out to $r = 2.79$, a fixed setting of the offline instrument: wide enough for the static throat and the collapse horizons, but an inflating throat can pass it, after which the edge reading bounds the throat’s areal radius from above. On the same spheres the null expansions $\theta_{\pm}$ are computed with the outward direction fixed by *increasing areal radius*, so that a dynamic throat is marginally *anti-trapped* [33]. The orientation matters: the in-code monitor fixes the outward direction by increasing *coordinate* radius, which points the wrong way beyond a throat and manufactures trapped verdicts whenever $R(r)$ falls. Every horizon statement here is therefore read offline from archived plotfiles with the areal orientation; the in-code columns serve only as a live monitor.

The star-shaped scan reports the outermost coordinate sphere that is trapped everywhere ($\max \theta_+ = 0$, $\theta_- < 0$). A trapped sphere lies inside the apparent horizon, so this sphere—“the scan’s MOTS” below—bounds the marginally outer trapped surface from inside. We quote its areal radius R and its Misner–Sharp mass [32],

$$\frac{2M_{\text{MS}}}{R} = 1 + \frac{R^2 \langle \theta_+ \theta_- \rangle}{4}, \quad (9)$$

with the product averaged over the sphere. On an exact MOTS $M_{\text{MS}} = R/2$; on the scan’s spheres M_{MS} exceeds $R/2$ by at most 3% about a lone collapsed throat, but about a merged remnant by up to 34% while the surface forms (44% at level 3), falling to 4% as it rounds. This is the shape systematic of star-shaped scanning, and it also affects whether a surface is found at all. Because the phantom violates the null energy condition, a trapped surface need not lie inside an event horizon; “horizon”, “censored” and “black hole” below refer to MOTSs on the

evolved slices. A deformable instrument backs the scan: a spectral flow finder expands the trial surface as $r = h(\theta, \phi)$ in real harmonics up to degree $\ell_{\max} = 6$ (8 where stated) and drives its expansion to zero by a damped fast flow, with the same orientation rule. It recovers Kerr–Schild Schwarzschild ($R = 2M$ to 0.24% and $M_{\text{MS}} = M$ to 0.12% from spherical and dented seeds), correctly finds no surface on the time-symmetric Ellis throat ($\theta_- \geq 0$ there), and reproduces the star-shaped scan’s surface on the head-on remnant.

Radiation is read from Ψ_4 on spheres at $R = 10, 14, 18$ (and 22 on the single-throat wave arms) on the head-on and the single throats, whose $L = 128$ arms keep these spheres, and at $R = 20, 28, 36, 44$ on the $L = 128$ spiral and fly-by, decomposed in spin-weight -2 harmonics. Every amplitude quoted is $r\Psi_4$; on the head-on the in-code and offline pipelines agree at the burst peak to 0.2%, while on the single throats the in-code stream is dominated by its grid-scale floor and only the offline one is used. A scalar stream is read on the same spheres, and at $R = 14$ and 30 on the $L = 128$ spiral and fly-by: ϕ and Π in spin-weight-0 harmonics, $A_{\ell m} = R\phi_{\ell m}$, and the kinematic flux $F_{\text{kin}} = -R^2 \oint \Pi \partial_r \phi d\Omega$, positive for an outgoing *canonical* wave (Sec. VIII F). A per-cell autopsy records the location and one-step history of any floating-point failure.

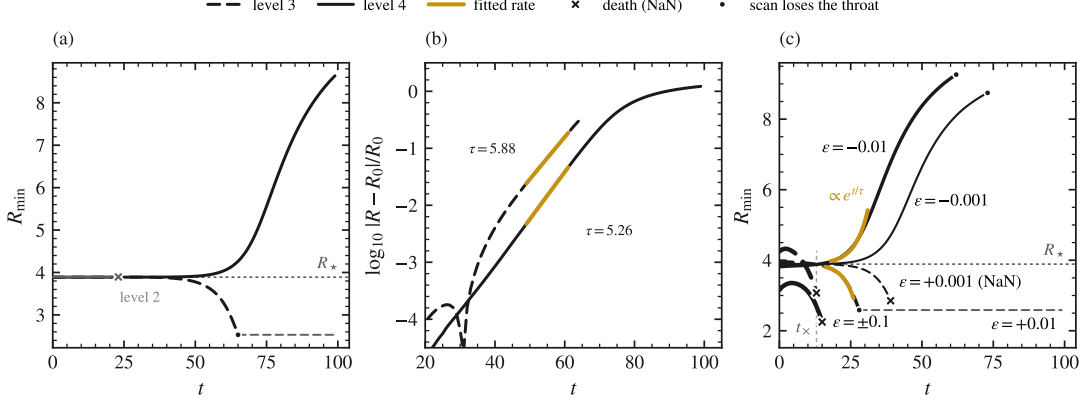


FIG. 1. Instability of the isolated throat. (a) Minimal areal radius from exact static data at refinement levels 2–4. The exact solution (R_* , dotted) is held until truncation noise excites the unstable mode: level 3 (dashed) collapses, level 4 (solid) inflates, level 2 (grey) dies of an origin failure first (cross). Dots mark where the scan loses the throat; muted lines carry the last reading. (b) Deviation from R_* : one exponential mode with a level-dependent sign, $\tau = 5.88/5.26$ at levels 3/4. (c) Level 3 with declared seeds, Eq. (7): $\varepsilon = \pm 10^{-2}$ (heavy), $\pm 10^{-3}$ (light) and, heaviest, ± 0.1 , which both collapse; dashed, outward kicks. Twins cross at $t_\times \simeq 13.0$ independent of ε . Gold: exponential fits; crosses: deaths at the compactified origin, $t = 40.1$ ($+10^{-3}$), 14.1 ($+0.1$) and 15.2 (-0.1). Their constraint norms are Fig. 10(a,b). Unless a seed is declared, the grid’s truncation noise picks the branch.

IV. THE SINGLE WORMHOLE

A. The static throat as a fixed point

Evolved without a declared perturbation, the exact drainhole holds $R_* = 3.8895$ to 0.1% for 35 time units at level 3 and to 1% until $t \simeq 44$ (Fig. 1(a)); at level 4 it is exact to four digits at $t = 30$. The departure moves *later* by 5.6–8.7 units per halving of the cell—a truncation-noise seed of fixed growth rate handed a smaller amplitude. Level 2 dies of an origin failure at $t = 24.2$ with the throat still exact to 0.12%. A lighter throat is less stable: at level 3 the $m = 0.5$ throat holds 0.1% only to $t \approx 17$, against 35 at $m = 1$, and dies at its origin at $t = 26.3$.

B. The unstable mode

The departure is a single exponential mode (Fig. 1(a,b)): plateau fits give $\tau = 5.88$ (level 3) and 5.26 (level 4), i.e. $T \equiv \tau \alpha_{\text{th}}/R_* = 0.87$ and 0.78, where $\alpha_{\text{th}} = e^{u(r_t)} = 0.575$ is the static lapse at the throat, so that T is the proper-time e-fold in units of the throat’s light-crossing time. Our drainhole is the $\gamma_1 = m/a = 0.5$ member of the static family of González, Guzmán and Sarbach (their metric with $b = a$, $\gamma_0 = -\pi\gamma_1/2$), for which their linear analysis gives $T = 0.758$; their Table I spans $T = 0.846$ at the massless end to 0.590 [15]. Solving the linearised field equations of this solution ourselves—their regularised master equation and, independently, the reduction in the gauge $\delta\phi = 0$ —gives $T = 0.7584$, i.e. $\tau_{\text{lin}} = 5.13M$. The measured e-fold approaches it with resolution: it is 2.5% longer at level 4 and 15% longer at level 3.

The seed is truncation error, and its sign flips with resolution—level 3 collapses, level 4 inflates—confirming from exact massive data the two branches found for the massless throat [44].

C. Branching: collapse versus inflation

A declared kick, Eq. (7), overrides the truncation seed (Fig. 1(c)). At $\varepsilon = \pm 10^{-2}$ and $\pm 10^{-3}$ the fate is always opposite to the push, each $\pm$ pair crosses at the ε -independent $t_\times \simeq 13.0$, and a tenfold smaller kick arrives 11–12 units later at matched horizon radius—linear response on the same mode. The branching survives refinement: at level 4, whose own seed inflates the unkicked throat, $+10^{-2}$ still forms a horizon at $t = 11$ and -10^{-2} still inflates, the twins crossing at $t = 12.99$ against level 3’s 13.01. The branch point is the throat’s, not the grid’s or the kick’s. The kick also sets the clock: horizons form at $t = 11$ ($\varepsilon = +10^{-2}$) and $t = 25$ ($+10^{-3}$), against $t \leq 61$ for the unkicked throat (its first scanned slice after departure already holds a MOTS). At $\varepsilon = \pm 0.1$ the data are no longer perturbative (Fig. 1(c)): $+0.1$ turns the minimal surface into a maximum between two minima and is trapped from the first scan ($t = 1$), while -0.1 , which narrows the throat to $R_{\text{min}} = 3.15$, re-expands to 3.36 by $t = 4$ and then collapses as well, trapped from $t = 6$ —against the sign rule. Both die at the compactified origin ($t = 14.1$ and 15.2, NaN in h_{11}), the failure that also ends the $+10^{-3}$ arm at $t = 40.1$, before the lapse has collapsed anywhere. Their Hamiltonian defect is $3\text{--}6 \times$ that of the 1% arms in the box norm and $8.6\text{--}10.3 \times 16\pi|\rho|$ on the shell; the usable range ends at 10^{-2} .

D. The collapse endpoint

The collapse branch ends in a black hole that consumes its supporting field. On the unkicked level-3 arm the horizon’s areal radius falls $3.23 \rightarrow 2.37$ and $M_{\text{MS}} 1.62 \rightarrow 1.23$ over 39 units, with the origin lapse at 0.016; the kicked $\varepsilon = +10^{-2}$ arm runs clean to $t = 100$ behind its horizon. Unlike the massless throat of Ref. [44], which bounced at $t \approx 4M$ and shed its horizon by $18.5M$, no arm here sheds a horizon. The pure-quadrupole arm ($\varepsilon_2 = 10^{-2}$, no radial kick) is fully instrumented (Fig. 2): a MOTS from $t = 33$ at $R = 3.80$, shrinking to 2.41 ($M_{\text{MS}} 1.90 \rightarrow 1.24$), while $|K|$ grows inward and the lapse pinches to zero at the centre.

The shrinking is physical; the regrowth that follows it in the scans is not. In spherical symmetry a massless phantom can only shrink a horizon: with the double-null metric $-2e^{-f} du dv + R^2 d\Omega^2$, $T_{uv} = 0$ and $T_{vv} = -(\partial_v \phi)^2$, the Misner–Sharp mass on a marginally trapped tube obeys $\partial_v m = 4\pi R^2 e^f (\partial_v \phi)^2 \partial_u R \leq 0$ and $\partial_u m = 0$, and the tube is timelike or null [65], so $R = 2m$ can only fall, as phantom accretion requires [66]. In the three arms kicked with $\varepsilon = +10^{-2}$ —at level 3, and at level 4 dressed with $\varepsilon_2 = 0.005$ and 0.05 —the horizon falls by 40% in areal radius to a floor at $t \approx 47$ ($R = 2.34$), and the scans then show it growing back by 9–11% (Fig. 11 in Appendix A). That regrowth is a numerical artefact: it follows a Hamiltonian-constraint violation that reaches the horizon at the floor time, it is the same in a purely spherical level-4 arm, and the pure-quadrupole arm, whose seed has no $\ell = 0$ defect, only shrinks (Fig. 2(a)). What the collapse branch establishes is the shrink: the throat becomes a black hole that swallows its own phantom support and loses 40% of its horizon radius doing so, in minutes even at $10^6 M_\odot$ (Table II). Hawking’s area law assumes the null energy condition [34, 70], which the phantom violates; this horizon runs it backwards.

E. The inflation branch

The inflation branch forms no trapped surface. The $\varepsilon = -10^{-2}$ throat, evolved in an $L = 512$ box at level 5 (Fig. 3), keeps growing to the end of the record, $\times 3.8$ by $t = 218$, and stays anti-trapped: the $\theta_l = 0$ horizon sits on the neck and $\theta_k = 0$ runs outward, to $R \approx 73$ by $t = 212$. Until $t \approx 40$ the neck grows at the Shinkai–Hayward rate in its own proper time, $HR_0 = 1.22$ (locally 1.0–1.3), against their massless 1.1 [13] and our linear mode’s 1.30. After that the $1+\log$ slicing freezes the lapse at the neck ($0.57 \rightarrow 0.015$), only ≈ 5 units of its proper time pass by $t = 218$, and its growth per unit t slows—a property of the slicing, not of the throat.

Its end state is open, and the throat is shedding its own support: the monopole scalar flux through every sphere grows quasi-exponentially over the quotable window (Sec. VIII G). Three endings remain: saturation at a larger radius, disfavoured because static drainholes

TABLE II. The drainhole clock in physical units: the production throat ($R_\star = 3.89M$) scaled by its ADM mass M . $\tau = 5.9M$ is the e-fold time (Sec. IV B); the last three columns are the measured times to a horizon for the two declared kicks, both outward ($\varepsilon > 0$, the collapsing sign; Fig. 1(c)), and an upper bound for truncation noise (Sec. IV C). Intermediate seeds follow $t_{\text{BH}} \simeq 11M + \tau \ln(10^{-2}/\varepsilon)$.

M	$R_\star$	τ	time to a horizon		
			$\varepsilon = +10^{-2}$	$+10^{-3}$	noise ($\leq$)
$30 M_\odot$	172 km	0.87 ms	1.6 ms	3.7 ms	9.0 ms
$10^3 M_\odot$	5.7×10^3 km	29 ms	54 ms	0.12 s	0.30 s
$10^6 M_\odot$	5.7×10^6 km	29 s	54 s	2.1 min	5.0 min
$10^9 M_\odot$	38 AU	8.1 h	15 h	1.4 d	3.5 d

are unstable fixed points; exponential inflation in proper time, which spherical evolutions find [13, 14] and our onset matches; or a turnover into collapse once too little phantom support remains, which a flux still growing when its reservoir is spent would force. That reservoir now has a number: outside the static throat the field carries a Misner–Sharp deficit of $R_\star/2 - m = 0.94$. Its spending is only bracketed. Over the window in which the exterior slicing is still quiet, the arms of Sec. VIII G shed 0.03 through $R = 18$ —a few percent of the store—while on this arm the $1+\log$ front reaches every sphere before the monopole grows, and the energy through the coordinate-60 sphere by the wall reads 4.1 kinematically but 0.04 with the geometric factor of Sec. VIII F applied—either side of the store. A downturn of $R_{\text{min}}(t)$ followed by a trapped surface would decide the ending; none occurs by $t = 218$, the growth holds at every sample, and the anti-trapped boundary advances at a steady $0.33c$, which Sec. X C turns into a population-level bound.

F. Lifetime and physical scale

At fixed m/a the solution is scale-free, and the measured e-fold is $\tau\alpha_{\text{th}} = 0.78\text{--}0.87 R_\star$ in proper time—one light-crossing of the throat. The lifetime is a logarithm of the seed: the horizon times of Sec. IV C set a decade’s cost at $14M \simeq \tau \ln 10$, and the unkicked throat’s places the level-3 truncation seed at $\varepsilon \gtrsim 2 \times 10^{-6}$. Table II converts these to physical units. No static throat of astrophysical mass survives: a stellar-mass drainhole is gone in milliseconds, a supermassive one in minutes, and only at $10^9 M_\odot$ does the lifetime reach days. The logarithm makes this robust: ten further decades of quiet initial data extend a $30 M_\odot$ throat from 1.6 to 22 ms. A foam-born pair must collide as it is born; otherwise each throat collapses or inflates on its own first (Sec. X A).

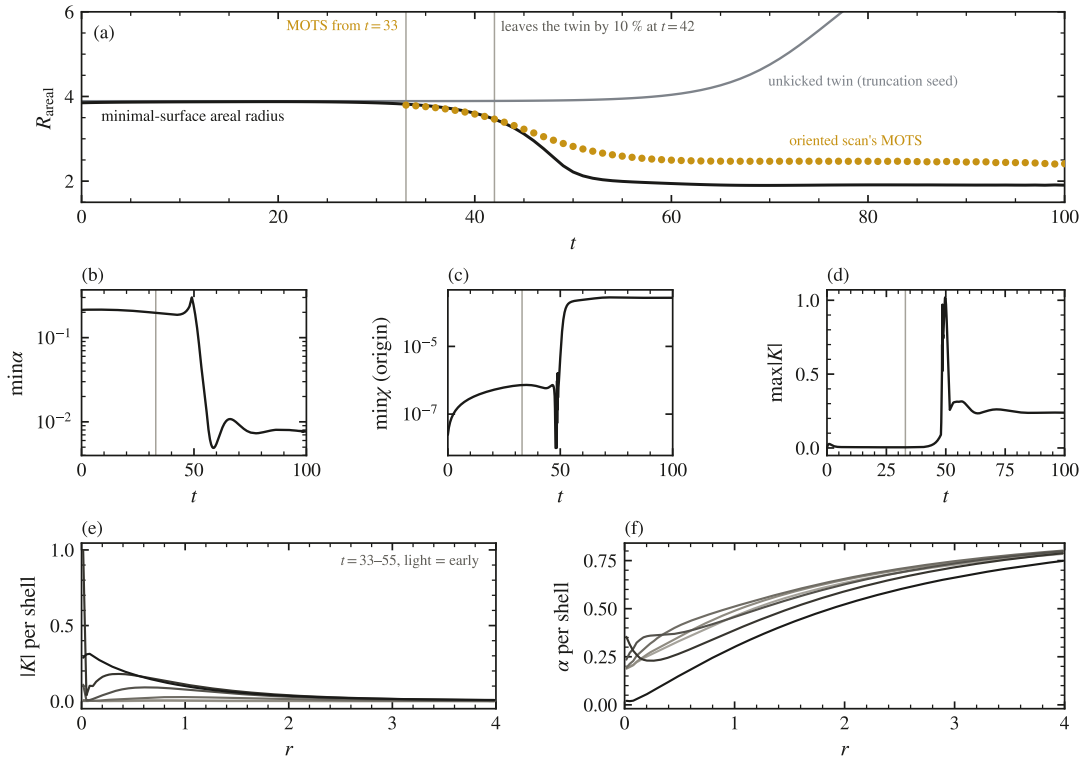


FIG. 2. Collapse of the pure-quadrupole arm ($\varepsilon_2 = 0.01$, no radial kick). (a) Minimal-surface areal radius (ink) and the scan's MOTS (gold) from $t = 33$: R 3.80 $\rightarrow$ 2.41, M_{MS} 1.90 $\rightarrow$ 1.24. Grey: the unkicked level-4 twin, which inflates on its own seed; this arm leaves it by 10% only at $t = 42$ —the second-order drive of the pure quadrupole. (b)–(d) Global $\min \alpha$, $\min \chi$ (the origin, the far side's compactified infinity) and $\max |K|$; the violent phase spans $t = 45$ –55. (e), (f) Radial profiles over 128 shells, $t = 33$ –55 (light = early): $|K|$ grows inward while α pinches to zero at the centre. Constraint norms: Fig. 10(c). The collapse ends in a black hole that consumes its supporting field.

V. TWO THROATS AT REST

A. The sign rule

Two identical throats released from rest push apart; reversing one throat's scalar, $\sigma = +1 \rightarrow -1$ in Eq. (6), turns the push into a pull (Fig. 4(a,b)). Both forces share one radial law, so their ratio Q is separation-independent: no gap, mass or approach speed lets a like-signed pair fall together, and $\sigma = -1$ is the only gravity-driven merger channel. The flip is not a tuning: $\phi \rightarrow -\phi$ leaves T_{ab} , and with it the one-body geometry, unchanged. The measured pull-to-push ratio at $a = 2$ is 1.518 ± 0.021 (scatter over 15 slices) against the $(Q+1)/(Q-1) = 3/2$ of Eq. (6), and the force follows the throat width—the scalar charge—not the mass, which is equal in all arms.

B. The force law

How the push falls with distance is read off a ladder (Fig. 4(c)). Four like-signed pairs, identical but for the gap $d = 12/14/16/18$, are released from rest, and by a common $t = 11.5$ their separations have grown by

$\delta d = 0.4696/0.3699/0.2980/0.2438$. For displacements this small, δd at a fixed time is proportional to the force, so under the inverse-square law of Eq. (6) $\delta d d^2$ would be the same on every rung. It is not: it rises monotonically, $67.6 \rightarrow 79.0$, across the ladder, which excludes inverse-square in the coordinate separation. Instead $F \propto (d+\delta)^{-2}$ with $\delta \approx 3$ –4 describes it: δ fitted on $d = 12$ –16 alone predicts 0.246 at $d = 18$, where inverse-square gives 0.209 and the measurement is 0.2438. Since δ is comparable to the placed pair's mouth radius (4.24 at $d = 12$; Sec. VC), the force is inverse-square in an *effective* separation exceeding the coordinate one by about a throat radius. Across widths, at $d = 12$, the push grows at every rung ($\delta d = 0.147/0.283/0.416/0.615$ by $t = 11$ for $a = 1/1.5/2/3$) with an effective exponent of 1.3–1.4; the point-charge a^2 scaling is not confirmed.

C. Placement systematics

Freshly placed pairs read +1.8% ($d = 48$) to +20.5% ($d = 6$) wider than the isolated throat, falling as $d^{-1.16}$ (Fig. 4(d,e)), so every same- d comparison here is made against that curve rather than against the lone throat.

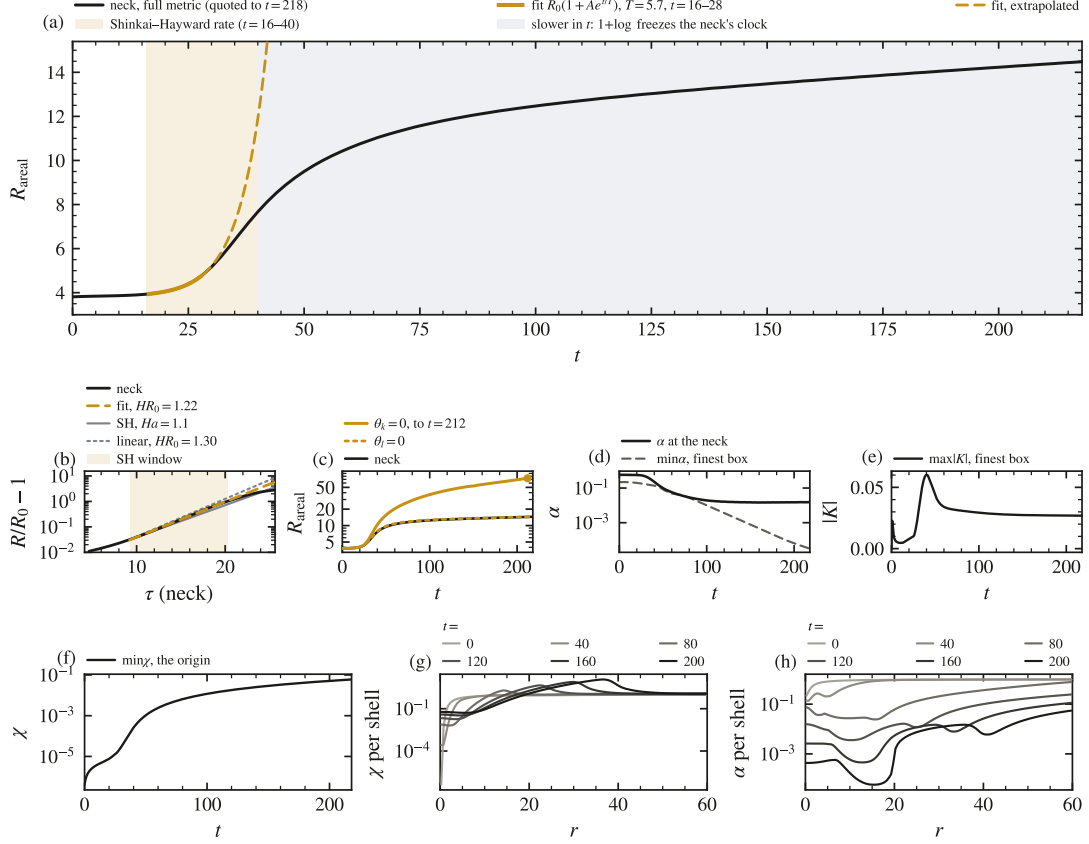


FIG. 3. The inflating throat ($\varepsilon = -10^{-2}$; $L = 512$, level 5) to $t = 218$. (a) Neck areal radius in the full metric, $3.81 \rightarrow 14.5$; gold, the exponential onset, $T = 5.7$ over $t = 16$ – 28 ; shaded, the Shinkai–Hayward rate ($t = 16$ – 40 , gold) and the slower growth in t once $1 + \log$ freezes the neck’s clock (grey). (b) Over that window the neck’s proper-time rate is $HR_0 = 1.22$ (local 1.0 – 1.3), against Shinkai–Hayward’s massless $Ha = 1.1$ and the linear mode’s 1.30 . (c) The $\theta_i = 0$ horizon stays on the neck; $\theta_k = 0$ runs out to $R \approx 73$ by $t = 212$. (d) The lapse at the neck falls $0.57 \rightarrow 0.015$; dashed, its minimum on the finest box. (e), (f) $\max |K|$ and $\min \chi$ on the finest box, around the origin. (g), (h) χ and α per shell (light = early): the χ peak moves out with the neck. Constraint norms: Fig. 10(d). The throat keeps growing, anti-trapped; its onset is at the linear rate.

Read that way, the released head-on pair’s mouths are genuinely squeezed, -0.6% at $t = 7$ growing to -4.0% by $t = 13$ —an interaction effect, since the lone throat holds its radius to 0.1% until $t \approx 35$.

VI. THE HEAD-ON MERGER

A. Contact and the common horizon

Released from rest— $p = 0$, no Bowen–York momentum on either throat—at $d = 8$ with opposite scalar signs, the pair reaches contact at $t \approx 20$ while both mouths are still open, and a common MOTS encloses them from $t = 22$: the corrected scan finds it at $r = 3.17$ with areal radius $R = 5.56$ ($M_{\text{MS}} = 2.99$; Fig. 5(a,b)). No scan finds a surface about either mouth at any time: the horizon is born common, and with the two throats’ area, $R = 1.01 \sqrt{2} R_*$. Its irreducible mass $R/2 \approx 2.8$ exceeds the initial data’s ADM mass of 2, a violation of the Pen-

rose inequality, which presupposes the non-negative energy density that the phantom lacks: each throat’s own Misner–Sharp mass, $R_*/2 = 1.94$, already exceeds its ADM mass of 1 by the negative energy of the field outside it. Formation is robust to the seed: an ensemble carrying $\varepsilon = \pm 10^{-2}$ on both throats—the amplitude that decides a lone throat’s fate—finds the same surface at $t = 22.0$, areal radius within 1.1% and mass within 0.5% , and the $-\varepsilon$ member tracks it to $t = 100$ within $\sim 2\%$ in areal radius and $\sim 1\%$ in mass, although the same kick moves the level-3 death by more than a unit ($t = 25.76$ against 26.91). The seed sets a lone throat’s fate and the grid’s lifetime, not the pair’s.

B. Through the wall to a black hole

Five units after the horizon forms, the level-3 scout dies at $t = 26.91$ with the classic anatomy— χ floored at the midpoint, $\max |K|$ running away to 209, a one-step

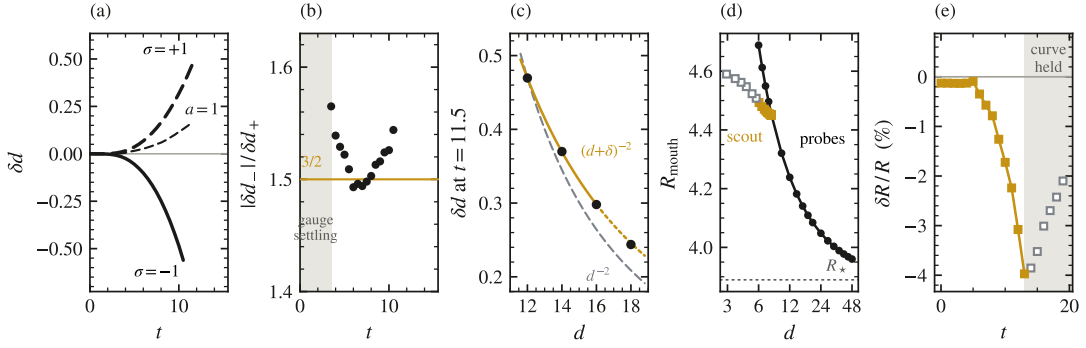


FIG. 4. Two throats at rest: the sign rule (a,b), the force law (c) and the placement calibration (d,e). (a) Separation change of three $d = 12$ pairs one knob apart: like-signed ($\sigma = +1$, heavy dashed) opens, opposite-signed ($\sigma = -1$, solid) closes, and the narrow like-signed pair ($a = 1$, light dashed) opens $2.8\times$ more slowly. Separations are inverse- χ -weighted centroids of the midplane χ pits. (b) Pull-to-push ratio against the prediction $(Q+1)/(Q-1) = 3/2$ (gold), where $Q = |F_\phi/F_{\text{grav}}| = (a^2 + m^2)/m^2 = 5$ is the scalar-to-gravity force ratio of Eq. (6): measured 1.518 ± 0.021 (scatter over 15 slices); grey, the gauge-settling window $t < 3.5$, excluded. (c) The force law: how far like-signed pairs released from rest at $d = 12/14/16/18$ have pushed themselves apart by $t = 11.5$ (dots), against inverse-square in the coordinate separation scaled through $d = 12$ (grey, dashed) and $A/(d+\delta)^2$ fitted on $d = 12\text{--}16$ alone (gold), dotted where it predicts the $d = 18$ rung (Sec. VB). If the push were inverse-square in d , every dot would lie on the grey curve; the dots rise above it, and the gold fit absorbs the excess with the offset δ . (d) Per-mouth areal radius of freshly placed pairs at $t = 0$ (circles), exceeding the isolated R_* (dotted) by an excess $\propto d^{-1.16}$. Gold squares: the head-on scout's mouths at the separation reached; open grey squares (here and in e) lie closer than the closest probe. (e) The scout against the placement curve, $\delta R/R = R_{\text{scout}}/R_{\text{place}} - 1$: the -0.13% plateau to $t = 5$ calibrates the tracker's grid snap, and the squeeze then grows to -4.0% by $t = 13$. In the shaded band the curve is held at its last point, so the magnitude is a lower bound; a boost can only raise a coordinate-sphere reading, so a reading below the curve is a genuine squeeze.

overflow. The failure is the grid's: at the same time the level-5 arm reads a smooth midpoint (lapse 1.6×10^{-2} , $\chi = 6.1 \times 10^{-8}$, $\max |K| = 0.11$) and runs to $t = 100$ with no restart or interior device, its Hamiltonian norm staying below its value at formation (Figs. 5(d) and 10(e)). The remnant is a black hole with both throats inside it, behind a common neck that is itself trapped, and it loses mass at every step as it swallows the phantom field (Fig. 5(a,b,e)): from $t = 36$, once the surface has rounded, M_{MS} falls on every scan row of the level-3 down-step arm, which tracks the horizon to $t = 100$, and of the three fill arms. By $t = 97$ the scan's areal radius is 4.17, a quarter below its value at formation: 57% of the two throats' area, $1.07 R_*$, and within 4% of $2M_{\text{ADM}}$, the Schwarzschild radius of the pair's mass. It falls on average by -0.0066 per unit, with a $\sim 1\%$ wobble as the scan's sphere follows a ringing surface, and no asymptote is resolved. M_{MS} levels off: through the down-step's rows $M_\infty + b e^{-(t-36)/\tau}$ gives $M_\infty = 2.160 \pm 0.006$ and $\tau = 19.4 \pm 0.8$ (a factor 4.8 better than a line), while other fit windows and the denser fill-arm scans give $M_\infty = 2.01\text{--}2.15$; since $2M_{\text{MS}}/R$ still falls from 1.23 to 1.04 over the window as the surface rounds, part of that levelling is the shape systematic of Sec. III C.

The horizon cannot bounce, and unlike the single-throat scans of Sec. IV D it never grows back: its mass falls at every row to the end of every arm that tracks it, and its radius, apart from that wobble, only falls. The reason is the spherical first law of Sec. IV D: on a marginally trapped tube a canonical field can only grow

a horizon, and the phantom can only shrink it. The head-on is not spherical, but its only positive-energy influx is gravitational radiation, and the whole event radiates $3.3 \times 10^{-3} M$, about 1% of what the horizon loses. The head-on thus shows the law without the numerical artefact that makes the single throat's scans regrow. The shrink ends when the negative energy outside is gone: the end state is the Schwarzschild hole of the pair's Bondi mass, $R = 2(M_{\text{ADM}} - E_{\text{GW}} - E_\phi) \approx 4.1$, a little above $2M_{\text{ADM}}$ because the scalar radiation carries negative energy away (Sec. VIII G), and by $t = 97$ the scan lies within a few percent of it. The late decline is not accretion: between $t \approx 45\text{--}55$ and $t \approx 90\text{--}99$ every proxy of the phantom flux into the horizon falls 15–180-fold, while the scan's rates fall only 3–4-fold. The spiral remnant ends at the same size (Sec. VII B), as it should if the end state is set by the pair's mass.

This arm reproduces an earlier level-3 $\rightarrow$ 5 restart chain—(2, 0) waveforms within 2.9–5.4% of peak over the burst, radiated energies within 3–8%—so the mesh seam carries no result. The exterior is also grid- and device-independent within that restarted family: against the level-3 fill twin, which shares its $t = 22$ restart state, the level-3 down-step arm agrees on the horizon mass to 0.24% and on the wave to 0.05–0.33% of peak over $t = 45\text{--}100$, and fill twins agree with no-fill arms to 0.01% of peak. After $t = 70$ the family and the level-5 arm from $t = 0$ part, by 4.5–62% of peak, the most on the outermost sphere.

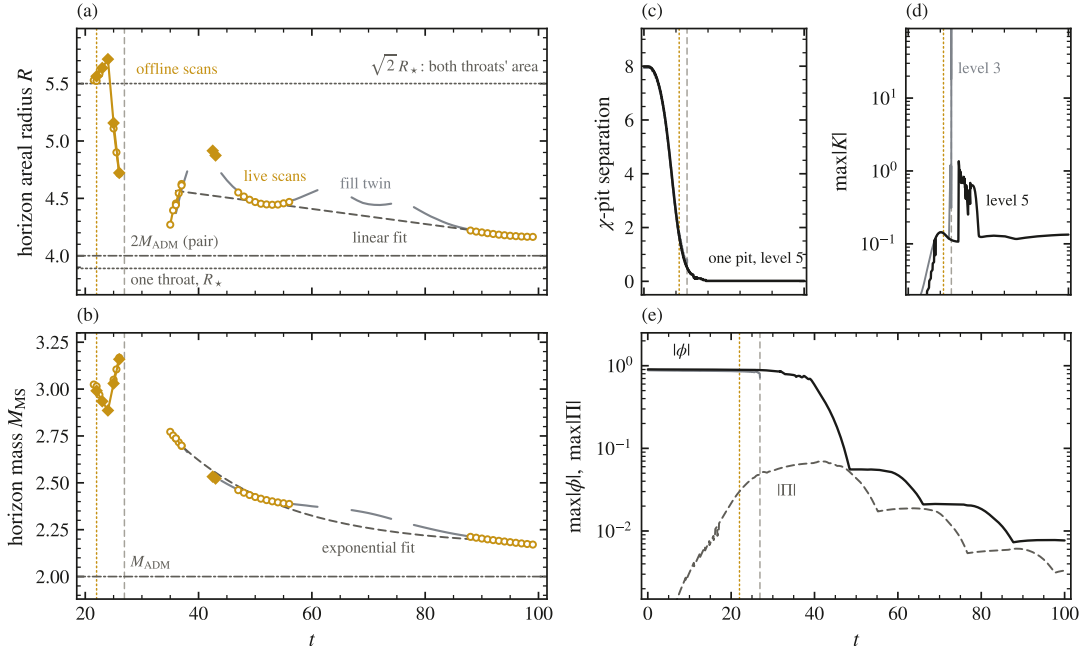


FIG. 5. The head-on merger ($d = 8$ from rest, $p = 0$, opposite signs) and the black hole it makes. Ink: the level-5 arm (level 5 from $t = 0$, no restart, no interior device, clean to $t = 100$); grey: the level-3 scout. (a) Areal radius and (b) Misner–Sharp mass of the scan’s MOTS: filled diamonds, offline fine scans (at formation on the scout, later on the level-3 $\rightarrow$ 5 restart chain); open circles, live scans (the level-5 arm until its scan’s aperture closes, then the level-3 down-step), each joined within a run of rows and never across a gap; grey line, the level-3 fill twin, whose scan sees through those gaps. Dashed: the fits through the down-step’s rows, a line (-0.0066 per unit) in (a) and $M_\infty + b e^{-(t-36)/\tau}$ in (b). Rules, the initial data: one isolated throat, $R_\star = 3.8895$; $\sqrt{2}R_\star$, a sphere with both throats’ area; $2M_{\text{ADM}}$, the Schwarzschild radius of the pair’s mass ($M_{\text{ADM}} = 2$ in (b)). The scan’s sphere lies inside the MOTS: while the surface forms it retreats inward, where M_{MS} reads high, and later R wobbles with the surface’s shape, while from $t = 36$ M_{MS} falls at every row. (c) Separation of the χ pits. (d) $\max|K|$: the scout runs away at its wall, the level-5 arm walks through. (e) Grid maxima of $|\phi|$ (solid) and $|\text{II}|$ (dashed): the field that held the throats open is swallowed. Hamiltonian norms: Fig. 10(e). Gold dotted rule: common-MOTS formation ($t = 22$); grey dashed rule: the scout’s death ($t = 26.91$).

C. The half-mass pair

Halving the throat mass moves everything earlier without avoiding the horizon. At $m = 0.5$, $d = 6$, the lapse falls from 0.44 to 0.038 over $t = 7.5$ –13, the tracker merges the mouths at $t = 11.1$, the offline scan finds a common MOTS at $t = 12$ ($R = 3.83$), and the run dies at $t = 14.0$ with the level-3 scout’s anatomy (Sec. VIB): χ on its floor, $\max|K|$ running away and a one-step overflow of the metric inside the common MOTS. The death belongs to the pair: the lone $m = 0.5$ throat is exact to 0.02% through $t = 14$ and fails only at $t = 26.3$, and differently—in K , at its compactified origin, the origin failure of Sec. IVA.

VII. THE SPIRAL MERGER

A. Merger or fly-by

Varying the tangential momentum p of Sec. IIC at $d = 12$ separates plunge from fly-by (Fig. 14 in Ap-

pendix B). Released from rest, the pair falls straight in, and every pair from $p = 0.12$ up to 0.25 plunges to a merged core: $p = 0.15$ repeats $p = 0.12$ ’s endgame (a plateau at pit separation 0.87 against 0.81, then the dive), and the pits of $p = 0.20$ and 0.25 close to 1.1–1.5 at level 3 before its wall. At $p = 0.35$ the pair passes at 2.75 and recedes to 4.0 by $t = 74$ (level 3, stopped there); $p = 0.45$ scatters. So every $p \leq 0.25$ merges and every $p \geq 0.35$ does not: the boundary between merger and fly-by lies between these two values, at 50–70% of the circular value under the combined pull. That pull, $1+Q$ times gravity’s (Eq. (6)), does not capture a pair by itself: it is central, so the pair keeps its angular momentum and swings past unless its closest approach falls inside contact. Nor do the fly-bys escape: both start below the circular momentum and so, in Newtonian terms, remain bound. What follows the pass is set by the throats, which inflate.

The $p = 0.45$ fly-by ($L = 128$, level 5 to $t = 100$) is a scattering pass: closest approach 4.8 at $t \approx 40$, back out to 9.6 by $t = 100$ and still separating. The turn is orbital—it comes before the mouths outgrow the scan window—but the recession after it is not that of two

compact throats. No trapped surface appears on the corrected scans at any time, and the run ends without a NaN. Instead both mouths inflate: from $t \approx 43$ the throat lies beyond the scan window ($r \leq 2.79$), whose edge reading—an upper bound on the throat’s areal radius—reaches 33 by $t = 97$, so from then on “separation” is that of the χ pits, which no scan ties to the throats (the faint stretch of Fig. 14). On the equatorial slices the pair’s scalar dipole is wound into a two-armed trailing spiral, and each mouth drives a crescent of $\chi > 1$ ahead of it ($\chi \sim 6$ by $t = 92$): the encounter ends not as two receding wormholes but as an expanding, orbit-wound cloud of the field that supported them.

B. Plunge and collapse

Every “spiral” here is a plunge: with $M_{\text{ADM}} \simeq 1$ per throat, the coordinate separation $d = 12$ is $d/M = 6$, the innermost-stable-circular-orbit (ISCO) value, and the throat tracks of the momentum scan (Fig. 14(a)) complete only 0.17–0.28 of a revolution before closest approach ($p = 0.20$ –0.35). Section X A shows that no wider pair could inspiral instead.

The production spiral ($p = 0.12$, $L = 128$) is evolved as a three-stage chain. Level 3 carries it from $t = 0$ through the plunge; the $t = 36$ checkpoint, taken before the pits meet, is restarted at level 5, which carries the merger to the curvature wall at $t = 60.45$ (Sec. VII C); and from that leg’s $t = 57$ checkpoint, when the common MOTS already encloses the core, the interior freeze of Sec. III B is armed at $r \leq 1.9$ and the exterior evolves on to $t = 100$. The freeze cannot reach the burst it records: its physical influence travels at most at light speed and arrives at the innermost extraction sphere only at $t \approx 75$, after the burst peak has passed there, and the gauge modes that outrun it carry at most 6×10^{-5} of peak (Sec. VIII C); a twin with the fill window moved changes the (2, 2) wave by at most 0.385% of peak (Fig. 13 in Appendix A); and over the 3.4 units in which the frozen and unfrozen level-5 arms coexist, their (2, 0) and (2, 2) modes agree to 0.013% of peak at the innermost sphere, to 5×10^{-13} of peak at the next and bit for bit at the outer two.

The production spiral falls from $d = 11.9$ until its two χ pits—each wormhole’s compactified far side—lie 0.31 apart, and collapses at the centre with both wormholes still open (Fig. 6). The merger’s constraint transient at $t = 41.2$ is the same at both resolutions—the level-3 leg reproduces the level-5 base-grid norms to a median 3.5%/0.7% (Fig. 10(f))—and the collapse after it is clean: the lapse falls to 2.0×10^{-3} , χ reaches its floor at $t = 58.4$, and $\max |K|$ grows $24\times$ over $t = 50$ –58 ($0.10 \rightarrow 6.1$ across the level-5 window) on a wall that lands on the throat; ϕ holds ± 0.89 while Π saturates at ± 4.1 .

The shape-free flow finder (Sec. III C) finds a common MOTS on every death-window slice of the two level-5 arms. On the production chain its areal radius is $R = 4.83$ at $t = 55.0$ ($\theta_- \simeq -0.05$, newly formed; con-

verged from an outer seed at $\ell_{\text{max}} = 8$), 4.80 at 56.0 and 4.77 at 57.0, 5.4 units before that arm’s NaN; on the arm evolved at level 5 from $t = 0$ it has $R = 4.72 \pm 0.01$ at $t = 59.0$, 0.94 before its NaN, bracketed by inner and outer seeds, 23% peak-to-peak in coordinate radius about the pit, with $\theta_- = -0.18$. On an exact MOTS $M_{\text{MS}} = R/2$: 2.41 at formation, 2.36 at $t = 59$. On the frozen-core arm’s live exterior (fill $r \leq 1.9$, inside the surface) the remnant at $t = 98$ –100 has $R = 4.15$ ($M_{\text{MS}} = 2.08$), still draining at 0.2% per unit, bracketed by a pointwise-trapped and a pointwise-untrapped surface. Both wormholes lie inside it. Their pits stay distinct until χ reaches its floor—at $t = 55$ –57.0 they are 0.34–0.31 apart, with χ 300–600 times higher between them—as they must, since no classical evolution changes the number of asymptotic ends [16]. Spheres enclosing one pit alone reach down to $R \approx 5.7$ –5.9, an upper bound on each mouth’s own throat, and from outside both are reached through one open neck, the smallest sphere enclosing the pair: $R = 4.10$ at $t = 55$ and 3.87 at $t = 60$, about one initial throat. For scale, a surface with both throats’ area would have $\sqrt{2}R_* = 5.50$: the newborn common MOTS is $1.24 R_*$, 0.77 of the two throats’ area and 13% smaller than the head-on’s at formation, and the remnant is $1.07 R_*$, like the head-on’s. At its death the remnant is therefore a black hole containing both wormholes behind a common neck: the head-on’s anatomy, reached through a plunge.

The throats grow while the orbit closes. A level-3 re-run carrying per-mouth scans measures each mouth growing $4.2385 \rightarrow 4.7567$ (+12.2%) by $t = 28$, the last time the two scan spheres are disjoint, while the separation closes $11.9 \rightarrow 3.6$ (Fig. 15 in Appendix B); the mouths agree to 2×10^{-4} , so this is the radial mode. Fitted over $t = 8$ –25, the growth excess $R/R_0 - 1$ has $\tau = 3.70$, against 4.39 in the fly-by and 5.3–5.9 in isolation: a companion shortens the throat’s clock, while the merging and scattering arms agree to 19%—the instability belongs to the throat, not to the encounter. The burst of Sec. VIII C is thus radiated by throats already a tenth larger than in the initial data.

C. The curvature wall

No grid carries the spiral through its merger. Restarted from one $t = 50$ checkpoint at maximum levels 3–7, every arm of the $p = 0.12$ run dies on an h_{11} NaN, and the gains per doubling saturate: +1.03, +2.51, +0.53, +0.06 units (Fig. 12(a) in Appendix A). Sixteen times the resolution buys 4.13 units, and halving the time step on level 5 buys +0.36, most of the level 5 $\rightarrow$ 6 gain. The core matter damping moves the wall by less than a unit. Starting refined does not help either: on the doubled box the $p = 0.12$ arm evolved at level 5 from $t = 0$ dies at 59.94, against 60.45 for the production chain restarted at level 5 from $t = 36$; a truncation seed two levels smaller buys nothing, so the grid does not seed

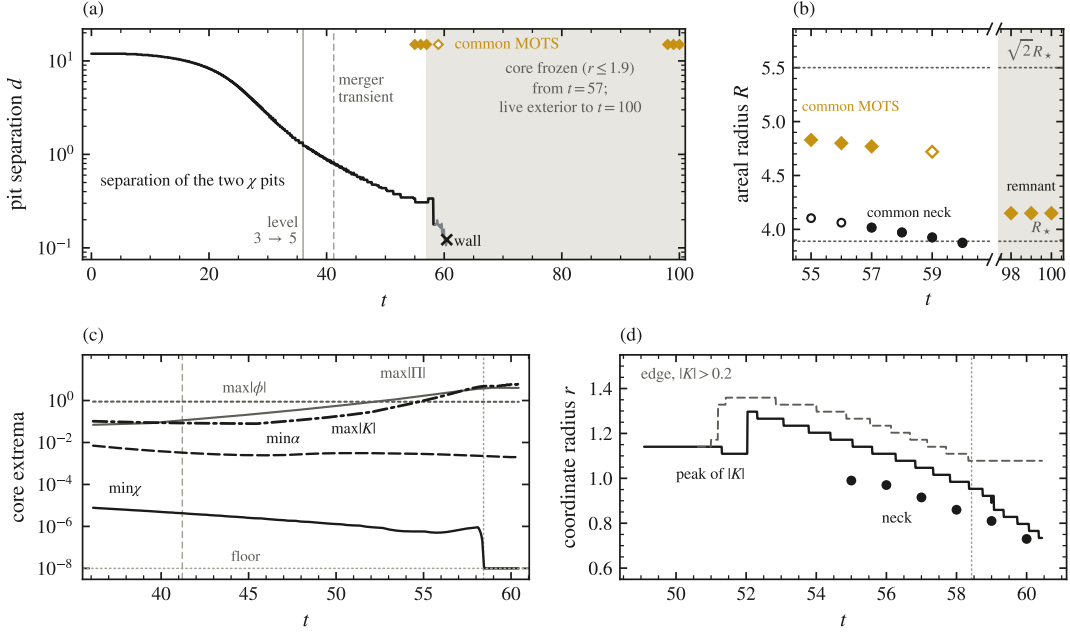


FIG. 6. The production spiral ($p = 0.12$, $d = 12$, $L = 128$) as one history: level 3 to $t = 36$, level 5 to the wall at $t = 60.45$ (cross), and the frozen-core leg (shaded: $r \leq 1.9$ frozen from $t = 57$, live exterior to $t = 100$), from which no core quantity is drawn. Rules: the level 3 $\rightarrow$ 5 handover (solid), the merger transient at $t = 41.2$ (dashed) and χ on its floor from $t = 58.4$ (dotted). (a) Coordinate separation of the two χ pits, each a wormhole's compactified far side, pale once χ is floored; gold diamonds mark the slices carrying a common MOTS. (b) Areal radius $R = \sqrt{A/4\pi}$ of the common MOTS (gold diamonds: filled, this chain; open, the arm evolved at level 5 from $t = 0$) and of the common neck, the smallest coordinate sphere about the pits' midpoint (circles: open, level 3; filled, level 5), against one initial throat (R_*) and the radius $\sqrt{2}R_*$ of the two throats' summed area. (c) Core extrema; $\max|K|$ dash-dotted. (d) Coordinate radius of the $|K|$ peak (from $t = 49.1$, before which the profile is flat), of its $|K| > 0.2$ edge (dashed) and of the neck (circles). Constraint norms: Fig. 10(f).

the wall. Past the wall the frozen-core stage (Sec. VII B) carries the exterior to $t = 100$.

The wall is censored. The spiral's common MOTS exists from $t = 55.0$ at the latest, 5.4 units before the production chain's NaN; the head-on's MOTS leads its level-3 death by 4.9 units; and lone collapses die behind their horizons at level 3 too ($\varepsilon = +10^{-3}$ at $t = 40.1$, 15 units after its horizon forms; $\varepsilon = +10^{-2}$ dressed with $\varepsilon_2 = 10^{-2}$ at $t = 27.6$, 14 after), while no level-4 collapse dies. Yet refinement rescues the head-on and never the spiral, although horizon presence, lead time and permanence are matched between them. Censorship is therefore necessary for a finer grid to pass a wall but not sufficient. Nor is the spiral's wall a stronger core at higher momentum, as far as $\max|K|$ can tell: across the momentum scan the level-3 wall arrives at $t = 52.1$ – 53.4 with no trend in p ; no level-5 arm shows a runaway in $\max|K|$ before its NaN, the $p = 0.15$ – 0.25 arms dying at $\max|K| = 0.71$ – 1.05 ; and the head-on passes a peak of 1.36 and survives. What falls with momentum is the grid's leverage: refinement from level 3 to 5 moves the wall by 3.5 units at $p = 0.12$, by under a unit at $p \geq 0.15$, and removes it for the head-on.

The natural suspect is the slicing. Every arm uses 1+log slicing with a Gamma-driver shift, whose slicings

can develop coordinate shocks [54] that refinement resolves rather than cures; a shock at a young, strongly deformed horizon would end every grid. Four gauge choices move the clock without removing the wall (Fig. 12(b); the three non-standard arms also carry the core matter damping, which alone moves the level-3 wall by under a unit). At level 3 the spiral dies at $t = 43.6$ with the 1+log coefficient halved, 49.0 with the harmonic-class driver $-2\alpha^2 K$, 52.1 in the standard gauge, and 61.9 with the shift damping raised $\eta = 1 \rightarrow 4$, which delays the whole sequence by ~ 10 units (the tracker merging the two centres at $t = 41.7$ against 32.1, χ floor at 55.6 against 44.9): the level-3 clock gains because the whole sequence is late—its wall still follows its χ floor by 6.3 units against 7.2—not because level 3 resolves more. On the head-on control the same knobs give deaths at 25.2 (harmonic), 26.9 (standard) and 34.2 ($\eta = 4$), and trapping precedes each: the harmonic arm's MOTS appears at $t = 20.8$ with $R = 5.47$ against the standard 5.40—the same surface on an earlier clock, 4.4 units ahead of the failure—and the $\eta = 4$ arm holds whole trapped spheres about the merged pit from $t = 30.8$, which bounds a MOTS from inside. The head-on's censorship is thus gauge-robust; only its clock moves. Refinement narrows the gauge spread without closing it: the $\eta = 4$ arm restarted at level 5 from $t = 50$ dies at

60.04, 1.9 units *before* its level-3 clock (a restart at $t = 60$ fails at once, at 60.05), while the standard gauge on the same box and restart dies at 55.60—a separation of ~ 4 units against 9.8 at level 3.

The wall is therefore grid-, seed- and device-independent within the Bona–Massó slicings and shift dampings tried: its clock is gauge-dependent, its existence is not.

VIII. GRAVITATIONAL WAVES

Every channel radiates, and the most energetic is the fly-by, which never forms a horizon. The peak $|r\Psi_4|$ of the dominant mode at the innermost sphere is 4.1×10^{-2} for the fly-by, 2.9×10^{-2} for the spiral, 2.3×10^{-2} for the head-on, 1.1×10^{-2} for the vacuum twin and 2.9×10^{-3} for the collapsing throat (Fig. 7). Each wormhole record propagates outward at c within its output cadence (Fig. 7, right); only the vacuum twin’s waveform lag exceeds its cadence, by 3.4 units across its two spheres, its inner sphere sitting in the near zone. No sphere is asymptotic: the dominant periods are comparable to the extraction radii, and the peak $r\Psi_4$ falls across the spheres by 20% on the head-on and 40% on the fly-by, whose outermost reading drops below the spiral’s. We therefore do not extrapolate, and carry the spread over spheres as the near-zone error bar of every energy (Fig. 8(d)).

The records are also clean of the initial data’s junk radiation. The seeded and superposed starts are not constraint-solved (Secs. II C and II D), but a defect radiating from $t = 0$ crosses an extraction sphere by $t \approx R$ —one light-crossing—while every quoted burst arrives with its physical trigger, tens of units later: the spiral’s merger transient at $t = 41.2$, the fly-by’s pass (all spheres peak at $t - R \simeq 34.5$), the head-on’s horizon at $t = 22$; the per-sphere envelopes rise with the burst, not at $t - R \approx 0$ (Fig. 7, right). The junk amplitude is measured where the defect is largest: the spherically kicked throat, whose shell defect peaks at $0.93 \times 16\pi|\rho|$ (Sec. II D), radiates nothing above the $\ell = 2$ numerical floor (Sec. VIII A), and the quadrupolar arms’ bursts are linear in ε_2 and unchanged when the radial kick is removed (Fig. 16)—the throat’s wave, not the shell’s transient. On the binaries the momentum constraint holds exactly at $t = 0$ and the Hamiltonian defect is $O(a^2/d^2) + O(m/d)$ (Sec. II C); its lasting effect is to excite the throats’ unstable mode, at an effective seed of 7×10^{-4} (Sec. X A).

In the dominant multipole the fly-by radiates $(4.2\text{--}7.4) \times 10^{-2} M$ across its spheres and the spiral up to $2.2 \times 10^{-2} M$, against $2.6 \times 10^{-3} M$ for the vacuum merger control—16–29 $\times$ and up to 8.5 $\times$ more. That control matches the spiral (grey in Fig. 7(c)); the fly-by’s own control is a vacuum binary at $d = 12$, $p = 0.45$. In vacuum that momentum is unbound (circular 0.204, parabolic 0.289), so the black holes start at periapsis and coast apart, radiating $1.05 \times 10^{-3} M$ in a single cycle as they swing away (grey in Fig. 7(d): peak $|r\Psi_4| =$

9.0×10^{-3} at the fly-by’s sphere, 4.5 $\times$ below the fly-by’s), whereas the drainhole pair falls from 12 to 4.8 under a pull six times gravity’s. The fly-by thus outradiates a vacuum binary of its own parameters by 40–70 $\times$.

A. Collapsing throat

A lone throat radiates only if its collapse is not spherical. The spherically kicked arm emits nothing above the $\ell = 2$ numerical floor, while the same kick with an $\varepsilon_2 = 0.05$ quadrupole produces a burst of peak $|r\Psi_4| = 2.9 \times 10^{-3}$, 35 $\times$ the control, carrying $3.2 \times 10^{-5} M$. The burst arrives in radius order at 0.8–1.0 c , carries the same $r\Psi_4$ to four spheres within 7.2%, and is linear in the seed (Fig. 16 in Appendix B): 4.85 $\times$ for a 5 $\times$ kick, ratios 1.99/1.97/1.87 at $R = 10/14/18$ for a halved seed, linear to 1–16% across $\varepsilon_2 = 0.005\text{--}0.05$. The pure quadrupole— $\varepsilon_2 = 0.01$ with the radial kick removed—radiates the same amplitude to 2%, its waveform leading the kicked arm’s by 1.1–1.5 M : the deformation sets the wave, the kick only its clock. These seed amplitudes follow the wave, not a decade grid: $\varepsilon_2 = 0.01$ matches the radial kick, whose usable range ends at 10^{-2} (Sec. IV C); 0.05 was added when the 0.01 burst stood only $\approx 2\times$ above the spherical control’s floor, too little to read a ringdown; and 0.005 halves the headline seed. A 10^{-3} arm would sit below that floor (Fig. 16(b)), and a 10^{-1} arm would deform the throat’s poles by $\approx 20\%$, as much as the $\varepsilon = \pm 0.1$ radial seeds that do not survive. The remnant rings at period 20.6 M (dashed in Fig. 7(a); an independent least-squares fit gives 20.61 ± 0.15), with no drift across the window (half-periods 9.4, 10.3 and 9.0 M), within a few percent of the Schwarzschild fundamental of its horizon’s Misner–Sharp mass over the late fitted window (19.5–20.3 M at $M_{\text{MS}} = 1.16\text{--}1.21$). It does not follow the larger mass the horizon carries at the window’s start (1.47, whose fundamental period is 24.7 M). The decay time is not measurable on this record.

B. Head-on ringdown

The head-on’s signal is a bell with no chirp: the (2, 0) breathing mode at $R = 10$ swings through +0.023, −0.018, +0.011, −0.006 at $t = 28.2, 43.8, 63.4, 82.2$ —a period of $\sim 31\text{--}39$, the amplitude falling $\times 0.6\text{--}0.8$ per half-swing—and arrives ~ 4 units later per 4 units of extraction radius (0.94 c), outgoing across all three spheres. On the detector track it is the bell that slows from 486 to 342 Hz in Fig. 8(c).

C. Spiral burst

The spiral’s (2, 2) burst peaks at $|r\Psi_4| = 2.9 \times 10^{-2}$ and folds onto one curve across $R = 20\text{--}44$ at $v/c = 1.00/1.00/1.00$. The burst is launched by the merger

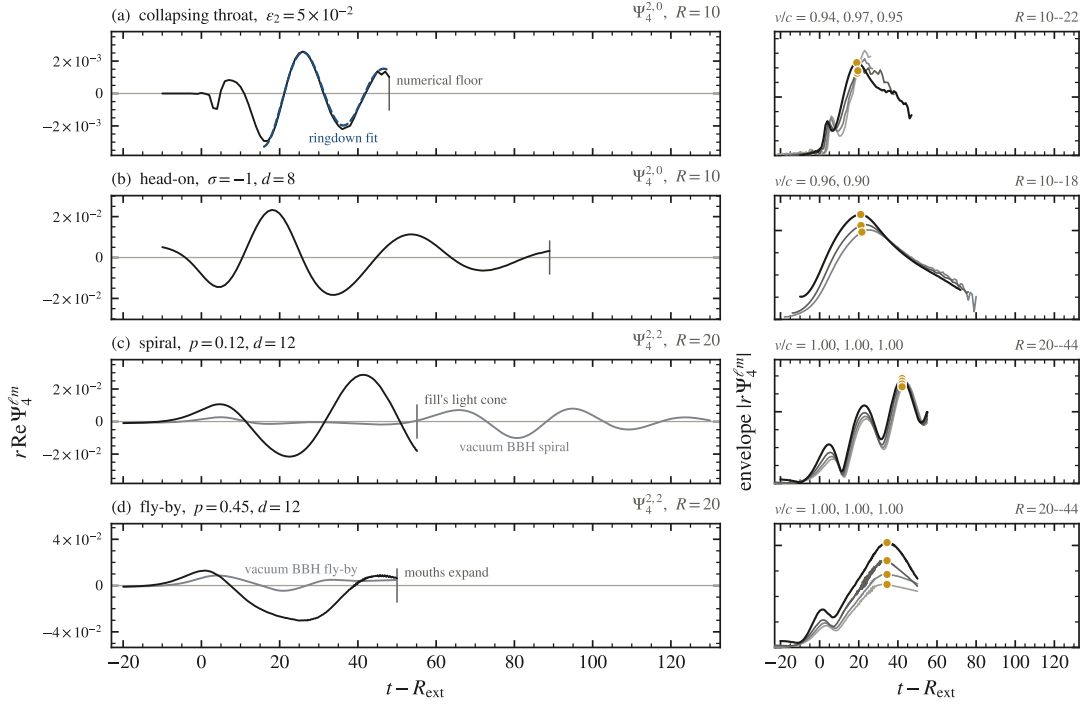


FIG. 7. The waveform gallery: one row per wormhole scenario on one retarded clock. Left: $r \text{Re} \Psi_4$ of the dominant mode at the innermost sphere, each row on its own scale; row (a) is the $\varepsilon_2 = 0.05$ arm, the only one loud and long enough to carry the ringdown, with the damped-sinusoid ringdown fit of Sec. VIII A dashed (period $20.6 M$). Caps mark where the drawn record ends: the collapsing throat where its numerical floor, the spherical control's own $r \Psi_4^{2,0}$, passes 10% of the burst peak; the spiral, carried from $t = 57$ by the frozen-core arm, at that core's light cone $t = 57 + (R - 1.9)$; the fly-by, complete to $t = 100$, at $t - R = 50$ on every sphere (Sec. VIII D). Grey in (c) and (d): the vacuum BBH spiral and fly-by, black holes with the same d and p (Sec. VIII), at the same sphere and on the same scale. The vacuum spiral's merger comes later and lower than the drainhole burst (Sec. VIII E); the vacuum fly-by, drawn over the fly-by's window, radiates one cycle as the holes swing away from periapsis and nothing at the pass, where the fly-by radiates its arch. Right: the amplitude envelope at every sphere on its own retarded time (ink to faint, inner to outer): $|r \Psi_4|$ for the complex $(2, 2)$ modes, the analytic-signal envelope for the real $(2, 0)$ modes, whose $|r \Psi_4|$ would fall to zero at each node. A signal moving at c and falling as $1/r$ folds onto one curve; each envelope rises with its burst (collapse, plunge, pass) and stops before a contaminant grows into it. Gold dots: the correlated wavefront. The quoted v/c is the lag of the whole complex waveform between neighbouring spheres over the drawn records, which, unlike peak timing, survives flat-topped envelopes (Sec. VIII E). Speeds: throat 0.94/0.97/0.95, head-on 0.96/0.90, spiral 1.00/1.00/1.00, fly-by 1.00/1.00/1.00.

transient at $t = 41.2$ and peaks at $R = 20$ at $t \approx 62$, before anything from the frozen core can arrive there; Sec. VII B and Fig. 13 bound the freeze's imprint on the later record.

D. Fly-by burst

The $p = 0.45$ pass emits a single $(2, 2)$ arch of peak $|r \Psi_4| = 4.1 \times 10^{-2}$ with no ringdown—there is no remnant—propagating at $v/c = 1.00$ across every sphere pair; in retarded time all four spheres peak together at $t - R \simeq 34.5$. The record is complete to $t = 100$ with no NaN, but is gated at $t = 70$, 15.5 units after the $R = 20$ peak: the mouths' expansion (Sec. VII A) contaminates the innermost sphere, whose $|r \Psi_4|$ stops decaying at $t = 76.1$, where the contaminant has grown to equal the decaying burst.

E. Vacuum twin

The vacuum control—same d , p and masses, no scalar—behaves as a binary black hole must: it merges, climbs the point-mass chirp $299 \rightarrow 473$ Hz and turns toward the equal-mass Kerr mode, its Ψ_4 peak at the textbook $fM = 0.065$. At the spiral's sphere the drainhole burst peaks $2.9\times$ higher than the twin's merger and 43 units earlier (grey in Fig. 7(c)). The twin also calibrates the wavefront method: its flat-topped merger envelope defeats peak timing, which reads $0.57c$ on it against the $0.82c$ of its whole-waveform lag, so every speed here is a correlated lag.

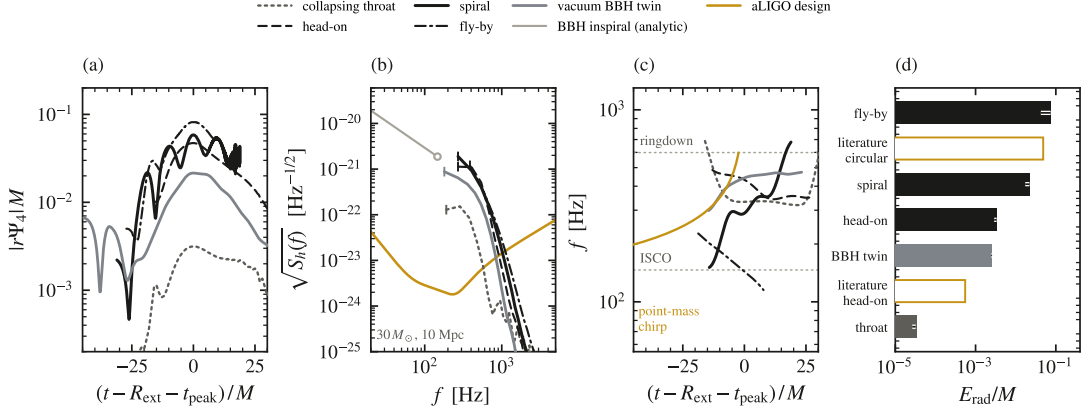


FIG. 8. Every source at one calibration: total mass $30 M_\odot$ at 10 Mpc, each record in units of its own total ADM mass. (a) Envelopes $|r\Psi_4|/M$ at the innermost sphere, aligned at each peak: fly-by loudest (8.2×10^{-2}) to lone throat (3.1×10^{-3}). (b) Strain amplitude spectral densities over the aLIGO design floor (gold), each from its own low-frequency corner (one cycle per record, ticked). The curves are falling power laws (slopes -3.4 to -5.8) with no resolved maximum, so any “peak” read here is the corner $1/T$. Faint line: the Newtonian inspiral of the same binary, $|\dot{h}| \propto f^{-7/6}$ to its ISCO at 147 Hz. (c) Instantaneous frequency against the point-mass chirp (gold), the ISCO (147 Hz) and the equal-mass Kerr ringdown (599 Hz): the vacuum twin climbs the chirp and peaks at $fM = 0.065$; no wormhole channel does (Sec. IX A). (d) Energy radiated in the dominant multipole; ticks give the spread over spheres (fly-by 43%, the widest). Open gold bars: published equal-mass vacuum values from rest at infinity—head-on infall 5.5×10^{-4} , quasi-circular merger 4.84×10^{-2} . Energies are band integrals of $|\int r\Psi_4 dt'|^2$ over a common retarded window, doubled for $\pm m$; gravitational channel only. At 10^4 – $10^6 M_\odot$ and $z_e \simeq 20$ the same records are millihertz LISA sources (Sec. XB, Fig. 9).

F. Scalar channel and its sign

Ψ_4 is blind to a spin-0 wave. Two opposite scalar charges—every merging pair here (Sec. V)—carry a dipole, where a vacuum binary’s leading moment is the quadrupole. The scalar stream rides the same spheres as Ψ_4 on the spiral and fly-by, so the two channels compare sphere by sphere (Fig. 17 in Appendix B). The recorded kinematic flux F_{kin} (Sec. III C) approximates the scalar energy crossing S_R per coordinate time. That energy is the flux, through the world tube that S_R sweeps out, of the matter energy current $J^\mu \equiv -T^{(\phi)\mu}_\nu t^\nu$ of the time vector $t^\mu = \alpha n^\mu + \beta^\mu$, with $T_{\mu\nu}^{(\phi)}$ the canonical stress tensor (the negative of Eq. (1)). Since $\nabla_\mu J^\mu = -\frac{1}{2}T^{(\phi)\mu\nu}\mathcal{L}_t g_{\mu\nu}$, the current is conserved where t^μ is a Killing vector and nearly so in the quasi-stationary exterior that the spheres sample, and Gauss’s theorem on the world tube gives its flux in 3+1 form [55, 56],

$$\begin{aligned} F_\phi^{\text{geo}} &= \oint_{S_R} \alpha s_i J^i \sqrt{\sigma} d^2x \\ &= - \oint_{S_R} \partial_t \phi (\alpha s^j \partial_j \phi + \beta_j s^j \Pi) \sqrt{\sigma} d^2x, \end{aligned} \quad (10)$$

positive for an outgoing canonical wave, with s^i the sphere’s outward unit normal, $\sqrt{\sigma}$ its area element, J^i the spatial components of the four-vector (not $\gamma^{ij}J_j$) and $\partial_t \phi = \alpha \Pi + \beta^i \partial_i \phi$. On a conformally flat sphere with uniform α and χ and no shift, $F_\phi^{\text{geo}} = \alpha^2 \chi^{-1/2} F_{\text{kin}}$; the shift adds an advective term $\propto (\beta^i \partial_i \phi)(s^j \partial_j \phi)$ and a term $\propto \beta_j s^j \Pi \partial_t \phi$ from its normal component. In the initial

data the sphere at $R = 30$ has $\alpha \approx 0.93$ and $\chi \approx 0.87$, and the area mean of $\alpha^2 \chi^{-1/2}$ is ≈ 0.94 , so energies built from F_{kin} overstate $|F_\phi^{\text{geo}}|$ by about 6%; the shift terms are neglected.

The channel is dipole-dominated and comparable to the gravitational one. On the fly-by at $R = 30$, $\ell = 1$ carries the scalar power $\sum_{\ell m} |\dot{A}_{\ell m}|^2$ to one part in 10^7 , the $\ell = 1$ signal travels outward at c from $R = 14$, and the time-integrated flux to $t = 60$ is $|E_\phi| = 4.3 \times 10^{-2} M$ —2.3 times the gravitational energy that has crossed the sphere by then (Fig. 17(a)). The ratio depends on the cut and on how E_{GW} is integrated. With the running integral of Fig. 17(a) it falls to 1.1 by $t = 70$ and 0.8 by $t = 80$, but that integral keeps growing after the burst has passed, accumulating the drift of the time-integrated Ψ_4 ; the band-limited energy of Fig. 8(d) levels off instead (2.7, 1.5 and 1.4 at $t = 60, 70$ and 80). The wave-zone mode sum gives 1.7, and the inner sphere returns 3.0 for the same window. We therefore use only the channel’s multipole, direction, sign and order of magnitude downstream.

Gravity couples to minus the canonical stress tensor, Eq. (1), so the physical energy flux is $-F_{\text{kin}}$ and an outgoing scalar wave carries negative energy; what the data establish is that the $\ell = 1$ wave is outgoing (on the fly-by, and on the head-on after its horizon forms) and of the size above. If the ADM balance holds with this term, a bound pair emitting it gains energy: the scalar channel pumps the orbit where gravitational radiation damps it, and radiation reaction is *anti-damped*. The same book-keeping appears at horizons, where swallowing the phantom shrinks the head-on remnant (Sec. VIB) and the

single-throat remnant (Sec. IV D).

G. Horizon censorship of the scalar channel

The head-on, re-run at level 5 from $t = 0$ with the scalar stream, is the one encounter that forms a horizon early, and the horizon switches the channel off (Fig. 18 in Appendix B). After the common MOTS closes ($t = 21.5$ on this arm’s live scan), the mouths’ remaining hair leaves as a single $\ell = 1$, $m = 0$ pulse, cresting later at each sphere in causal order, and the envelope then decays: log-linear fits over $t = 30$ –95 give $\tau = 19/24/29$ at $R = 10/14/18$, comparable to the $\tau = 19.4$ on which the horizon’s M_{MS} levels off in the down-step fit (Sec. VIB). The post-horizon energies keep the sign of Sec. VIIF, $E_\phi = -0.028/-0.036/-0.038 M$. Only this window is integrated: before $t = 22$ the flux at $R = 10$ runs canonically ingoing, the two open mouths’ static hair superposing.

Without a horizon the channel does not shut. The fly-by’s dipole envelope is still growing at the end of its record. The spiral’s horizon, present from $t = 55.0$, is only five units old at its wall, against a shedding clock of ~ 20 , so it is too young to show the shut-off.

The lone throat shows the same contrast. Re-run at level 4 with the scalar stream, the $\varepsilon = +10^{-2}$ throat collapses on schedule—a permanent MOTS from $t = 11$, R settling $3.88 \rightarrow 2.54$ —and its scalar envelope, after the burst, falls by 30 – $100\times$ across the spheres by $t = 80$, with late e-fold times of 6 – 7 . The unkicked level-4 throat, which inflates with no trapped surface by scan or flow finder, instead grows quasi-exponentially in the $\ell = 0$ monopole (e-fold ≈ 5 , shortening across $t = 45$ – 75), identically in a box-doubled ($L = 128$) copy through $t \approx 80$, so the growth is the throat’s and not the boundary’s or the sponge’s; beyond $t \approx 80$ the constraint norms grow, and only $t \lesssim 80$ is counted. The contrast is confounded with the fate: the horizonless arms are also those whose mouths inflate toward the spheres. What is measured is that the scalar channel shuts off behind a horizon and grows while a throat inflates.

IX. PATTERN MATCHING AGAINST DETECTOR DATA

In 2.26 h of coincident Advanced LIGO O3b data [30], a matched-filter search [27, 29] with 127 templates—the waveforms of Sec. VIII rescaled as $h \propto M/D$, $f \propto 1/M$ —finds *no candidate at zero lag*, and none was expected. No static throat of astrophysical mass survives (Sec. IV F), so LIGO could catch only the burst of an encounter or collapse at formation, never a wormhole in residence; and a primordial population forms and converts at $z \gtrsim 20$, far beyond the search’s reach and inside LISA’s band instead (Sec. XB). At single-detector SNR 8 and optimal orientation the median horizon over each

mass ladder is 4.8 Gpc (spiral), 2.5 (fly-by), 1.8 (head-on), 1.3 (vacuum twin) and 0.15 (collapsing throat); the first three are upper bounds, since their strain, from fixed-frequency integration [28], ends at 0.31–0.42 of peak.

The null result is not a blind spot. Cut to the signal’s duration, binary-black-hole templates (IMRPhenomD) match the bursts with fitting factors of 0.82–0.98, as they match the vacuum control (0.93–0.95), so burst morphology alone cannot tell the two apart. All 15 injections into O3b strain at optimal SNR 20 are recovered at 93–139% of optimal within 3 ms; only the lightest fly-by is demoted, by the χ^2 veto.

A. The frequency track

The frequency track separates a drainhole burst from a binary black hole (Fig. 8(c)). The vacuum twin peaks at $fM = 0.065$ and climbs the point-mass track before turning toward the equal-mass Kerr mode at $fM = 0.0885$: a monotone rise, then a fixed ringdown frequency. No drainhole channel does this. The head-on’s frequency *falls*, and mass loss cannot explain it: a black hole shrinking as its remnant does rings faster, by $\times 1.38$ over the same window, while the measurement is $\times 0.70$. The fly-by’s frequency falls as the pair separates; the spiral’s late record is carried by a frozen core and climbs without settling; the lone collapse rings at 0.55 of the equal-mass Kerr mode. A catalogue event [31] whose post-merger frequency holds within a few percent of a Kerr value therefore excludes the head-on channel at that mass.

B. Energy per event

The detector-independent deliverable is the energy per event in throat-mass units, in the dominant multipole (sphere spread in parentheses): fly-by $7.4 \times 10^{-2} M$ (43%), spiral $2.2 \times 10^{-2} M$ (21%), head-on $3.3 \times 10^{-3} M$ (13%), collapsing throat $3.2 \times 10^{-5} M$ (18%), vacuum control $2.6 \times 10^{-3} M$ (5%). Since Ω_{GW} scales as event density times energy per event, these convert any bound on Ω_{GW} [25, 26] into a bound on the abundance of primordial drainhole encounters, provided the ghost scalar exists at that epoch. The scalar channel adds comparable energy of opposite sign (Sec. VIIF).

X. DISCUSSION

All four fates of Sec. I occur, each with a measured selection rule: the relative scalar sign separates scattering from merger (Sec. V), the seed’s sign separates collapse from inflation (Sec. IV), and the orbital momentum separates plunge from fly-by at 50–70% of circular (Sec. VII). For detection, three results combine: no inspiral exists (Sec. XA), the wormhole phase is a burst whose frequency falls or drifts where a Kerr remnant’s would hold

(Sec. IX A), and the scalar counterpart decays once a horizon forms (Sec. VIII G). A surviving wormhole signature is a drifting-frequency burst with a scalar counterpart; a settled ringdown without one is already the descendant.

A. No wormhole inspiral

A drainhole binary cannot radiate an inspiral, and the obstruction is a logarithm. In a binary the mouths' unstable mode runs at $\tau = 4.39$ (fly-by) and 3.70 (merger; Sec. VII B), faster than the isolated 5.3–5.9; we use $\tau \simeq 4.4$. A quasi-circular pair at separation d merges by radiation reaction in $t_{\text{insp}} = \frac{5}{256}(d/M)^4 M/\eta$ (M the total mass, $\eta = 1/4$): 202 units from the ISCO, $d/M = 6$, and 640 from $d/M = 8$ —1.1 and 2.25 orbits at the initial period $T_{\text{orb}} = 2\pi\sqrt{d^3/M}$, which understates the cycles completed and so is conservative. That gives the instability $t_{\text{insp}}/\tau = 46$ e-folds at the ISCO and 146 for a two-orbit inspiral: the mouths reach merger unchanged only if the initial perturbation is below $e^{-46} \simeq 10^{-20}$, respectively 10^{-63} (with the isolated clock, 10^{-15} at the ISCO). Turned around, the lifetime is $\tau \ln(1/\varepsilon)$: seeds of 10^{-6} , 10^{-10} and 10^{-15} buy 0.33, 0.55 and 0.82 of one orbit. The orbit's demand grows as $(d/M)^4$ and the throat's endurance as $\ln(1/\varepsilon)$, so no finer grid or quieter data closes the gap: constraint-solved initial data, lowering the seed from the superposition defect's 7×10^{-4} to the level-3 truncation floor's 2×10^{-6} , are worth at most $\tau \ln 350 \simeq 26$ units, 0.14 of an orbit.

The inspiral time above is the vacuum quadrupole result, and this binary violates its premises. Its leading radiative moment is a scalar dipole carrying energy comparable to the gravitational channel's (Sec. VIII F). Were that energy positive, a reaction two to four times stronger would shorten t_{insp} and relax the ISCO requirement to 12–23 e-folds ($\varepsilon \lesssim 10^{-5}$ – 10^{-10}). Its energy is negative, however, so if the ADM balance holds the orbit is pumped rather than drained and no quasi-circular decay exists to compute. A wormhole inspiral fails on the throat's clock and again on the orbit's.

The wormhole-specific signal is therefore a short transient: within a few tens of M each mouth collapses or inflates, and the plunge, burst and conversion of Secs. VI–VIII are all that this source class emits *as wormholes*. Three qualifications apply. The argument covers a minimally coupled ghost scalar without spin, the matter model of the González-Guzmán-Sarbach analysis, and holds only while the throat's unstable mode exists. Rotation can remove that mode: Ellis wormholes set spinning exist up to an extremal-Kerr limit [57]; in five dimensions, with equal angular momenta, the static throat's unstable mode disappears above a critical angular momentum [58]; in four, radial perturbations to second order in the rotation indicate the same [59, 60], and the non-axisymmetric modes computed at rapid rotation show no instability [61]. Other supports are untouched by the argument and include radially stable throats—thin

shells for suitable equations of state [62] and dilatonic Einstein–Gauss–Bonnet wormholes, which need no exotic matter [63]—so a spinning or differently supported pair that outlives an inspiral is not excluded here. “No inspiral” means no *wormhole* inspiral: any long chirp belongs to the objects the throats become. And the seeds here are numerical, which the argument tolerates because its seed dependence is logarithmic and because the superposition defect's excitation of the mode is measured: back-extrapolating the fly-by's mouth growth gives an effective seed of 7×10^{-4} , just below the declared-kick decade over which the response is measured linear (Sec. IV C).

B. Heavy seeds and LISA bursts

Collapse, alone or in a merger, is a seeding mechanism. A throat converts to a black hole in seconds at 10^4 – $10^5 M_\odot$ and in minutes at $10^6 M_\odot$ (Table II), and the remnant keeps a mass of order its progenitors': the head-on and spiral remnants both end at $R \simeq 4.16$ (irreducible mass 2.08) from pairs of ADM mass 2, and the lone collapse at $R \approx 2.3$ from a throat of mass 1 (Secs. IV D, VIB, VII B). The scalar hair leaves on a $\sim 20 M$ clock after horizon formation (Sec. VIII G), so the remnants then accrete as ordinary black holes. From 10^4 – $10^5 M_\odot$ at $z \simeq 20$, Eddington-limited growth (Salpeter e-fold $\simeq 45$ Myr) reaches $10^9 M_\odot$ in 9–12 e-folds, 414–520 Myr, inside the ~ 600 Myr available by $z \simeq 7$: no super-Eddington phase is needed, which is what defines a heavy seed [52] (Fig. 9(a)). The channel could thus seed JWST's overmassive black holes and little red dots [46–49] without an atomic-cooling halo, a critical Lyman–Werner flux, or any baryonic environment.

Halo independence is itself a signature: the seeds pre-date baryonic structure, so M_{BH}/M_* starts effectively infinite and relaxes downward, making UHZ1's overmassive morphology the generic expectation rather than a tuning. The abundance required is small: one seed per present-day massive galaxy is $n \simeq 10^{-4}$ – 10^{-2} Mpc^{-3} comoving, a dark-matter fraction $f = nM/\rho_{\text{DM}} \simeq 3 \times 10^{-11}$ – 3×10^{-7} at 10^4 – $10^6 M_\odot$. And the channel evades the bound that closes the conventional route: FIRAS μ -distortions exclude primordial black holes of 10^4 – $10^{13} M_\odot$ formed from Gaussian curvature perturbations at any cosmologically relevant abundance [53], but these black holes form by wormhole collapse, not from enhanced curvature perturbations—the evasion the bubble and domain-wall channels also use [5, 6].

The encounters radiate, which makes the channel falsifiable, and the detector is LISA (Fig. 9(b)). Each burst carries the energies of Sec. IX B at $f_{\text{obs}} = (fM)_{\text{peak}}/[M(1+z_e)]$; with the measured $(fM)_{\text{peak}} = 0.03$ – 0.06 the band is 0.3–60 mHz across 10^4 – $10^6 M_\odot$ at $z_e \simeq 20$: LISA's band, not LIGO's. One at a time the bursts are loud. Rescaled with the strain of Sec. IX to $z_e = 20$ in a Planck cosmology [69] and filtered against the analytic LISA noise [67], the fly-by and the spiral

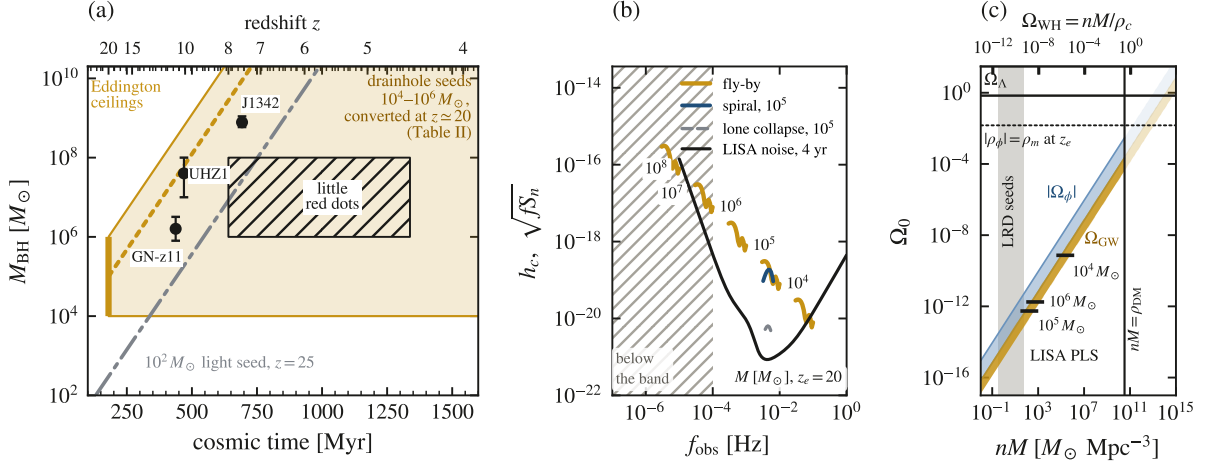


FIG. 9. The heavy-seed channel from measured numbers. (a) Seed growth: a drainhole population of 10^4 – $10^6 M_\odot$ converts at birth ($z \simeq 20$, seconds to minutes; Table II), so its reachable region (gold) runs from the lightest seed idling to the heaviest seed’s Eddington ceiling (Salpeter e-fold 45 Myr). UHZ1 [46, 47], GN-z11 [50], the $z = 7.5$ quasar J1342+0928 [51] and the little-red-dot box [48, 49] lie inside it; the first three lie above the ceiling of a $10^2 M_\odot$ light seed started at $z = 25$ (grey), which misses UHZ1 by two decades. (b) One burst against LISA: the characteristic strain $h_c = 2f|\tilde{h}|$ of a fly-by (gold) at each mass decade from 10^4 to $10^8 M_\odot$ (exponents at the tracks), and of a spiral (blue) and a lone collapse (grey) at $10^5 M_\odot$, all at $z_e = 20$, against $\sqrt{f}S_n$ of the analytic LISA noise [67] with the four-year galactic confusion (ink); inclination averaged, each track drawn over the band holding 90% of its $\int h_c^2 d \ln f$; the $10^7 M_\odot$ fly-by’s counted signal lies in its tail above the band floor, past the drawn core. The squared ratio of track to noise, integrated over $\ln f$, is the burst’s squared signal-to-noise ratio, so a short track can sit above the noise and still fall short of 8, as the lone collapse does. Heavier bursts are louder but redder: the tracks climb a decade per mass decade, the noise climbs faster below 0.1 mHz (hatched: below LISA’s band, not counted), and the $10^8 M_\odot$ burst lies wholly below it. At the abundances of (c) the population is ~ 0.05 –5 resolved bursts per year (Sec. X B). (c) Ω_{GW} of the conversions (gold, E/M from the head-on to the spiral) and the negative scalar deposit $|\Omega_{\phi,0}|$ (blue, $|E_\phi| = 0.8$ – $3.2 E_{\text{GW}}$ with E_{GW} from the spiral to the fly-by: every encounter, the case most favourable to Λ) today against the comoving mass density nM of converting wormholes, on which alone both depend (top axis: $\Omega_{\text{WH}} = nM/\rho_c$). Solid up to the dark-matter ceiling $nM = \rho_{\text{DM}}$ (vertical rule), faint beyond it: Ω_Λ is reached only at $\Omega_{\text{WH}} \simeq 60$ –800, above the cap $|\rho_\phi| = \rho_m$ at z_e that $H^2 > 0$ imposes (dashed). Ink ticks: the four-year power-law-integrated LISA sensitivity [68] at each mass’s conversion frequency $(fM)_{\text{peak}}/[M(1+z_e)]$. Grey: one seed per little red dot [48, 49] at 10^4 – $10^6 M_\odot$.

merger, the two loudest channels of Sec. VIII, reach optimal signal-to-noise ratios of 175–500 at 10^5 – $10^6 M_\odot$, and the head-on 61–76. All three stay above 8 from 3×10^4 to $4 \times 10^6 M_\odot$ with the inclination averaged, the galactic confusion noise added and only each record’s resolved band counted; the strain pedestal of Sec. IX matters only at the light end. Heavier events are louder but redder, and LISA’s low-frequency noise rises faster than their strain (Fig. 9(b)): at 10^7 – $10^8 M_\odot$, where the bursts peak at 3–60 μHz , only the fly-by, whose spectrum extends furthest above its peak, still reaches 8, up to $2 \times 10^7 M_\odot$. The spiral and head-on bursts each announce a seed, since each ends in a black hole; the fly-by, the loudest, leaves two inflating mouths, announces none, and occurs at a rate not tied to the seeds’. A lone collapse, radiating only $3 \times 10^{-5} M$ (a background of 10^{-16} – 10^{-14}), stays below detection on the same count, at a signal-to-noise ratio of at most ≈ 6 (≈ 10 only at optimal orientation). What limits detection is occurrence: every seed converts once, so LISA records $4\pi D_c^2(z_e) c T n \simeq 1.9 \times 10^3 (n/\text{Mpc}^{-3})$ bursts in $T = 4$ yr, ~ 0.05 –5 per year for $n = 10^{-4}$ – 10^{-2}Mpc^{-3} , each

$\sim 400 (M/10^5 M_\odot)$ s long. Averaged over time the conversions give $\Omega_{\text{GW}} \simeq n E_{\text{rad}}/[\rho_c(1+z_e)]$: with $E_{\text{rad}}/M = 3.3 \times 10^{-3}$ (head-on) to 2.2×10^{-2} (spiral) at $M = 10^5 M_\odot$, $n = 10^{-4}$ – 10^{-2}Mpc^{-3} gives $\Omega_{\text{GW}} \sim 10^{-14}$ – 10^{-11} at ~ 6 mHz, above LISA’s four-year power-law-integrated sensitivity [68] ($\Omega \sim 10^{-13}$ – 10^{-12} near millihertz [22, 24]) for $n \gtrsim (2\text{--}7) \times 10^{-4} \text{Mpc}^{-3}$ at 10^6 – $10^5 M_\odot$ if every conversion radiates like the spiral (Fig. 9(c)); fly-bys, which convert nothing, add up to $7.4 \times 10^{-2} M$ each at a rate the evolutions do not fix. At these rates, though, the signal is resolved bursts, never a stochastic background. The channel is therefore detectable by LISA for merger conversions of 3×10^4 – $4 \times 10^6 M_\odot$ with $n \gtrsim 5 \times 10^{-4} \text{Mpc}^{-3}$, where at least one burst falls in four years, and not for lone collapses, for $10^4 M_\odot$ seeds or for $n \lesssim 10^{-4} \text{Mpc}^{-3}$. The level and band scale as $1/(1+z_e)$, and $z_e \simeq 20$ is about the latest conversion epoch the seed arithmetic allows; a population converting at horizon re-entry would lie far below any band. Its spectrum is a pure burst population with no $f^{2/3}$ inspiral ramp (Sec. X A), unlike any direct-collapse scenario, whose seeds form hydrodynamically.

The same events deposit negative energy. The encounters radiate scalar energy comparable to their gravitational energy, with negative sign (Sec. VIII F), and an inflating throat sheds its support as a slower monopole of the same sign (Sec. IV E), so a primordial population fills the early universe with a diffuse negative-energy scalar background. It cannot be the dark energy, for three independent reasons (Fig. 9(c)). Its equation of state is wrong: massless waves have $w = 1/3$ and dilute as a^{-4} , losing a factor $(1+z_e)^4 \simeq 2 \times 10^5$ against Λ by today, and with $V \equiv 0$ no configuration does better, since the averaged $w = (\frac{1}{2}\dot{\phi}^2 - \frac{1}{6}|\nabla\phi|^2)/(\frac{1}{2}\dot{\phi}^2 + \frac{1}{2}|\nabla\phi|^2)$ of a massless field of either sign lies between $-1/3$ and $+1$ —so the inflating branch’s monopole cannot supply $w \approx -1$ either. Its sign is wrong: $\rho_\phi < 0$ lowers H^2 where a positive Λ raises it. And its size is wrong: $|\Omega_{\phi,0}| = n|E_\phi|/[\rho_c(1+z_e)]$ reaches $\Omega_\Lambda \simeq 0.69$ [69] only at $\Omega_{\text{WH}} \equiv nM/\rho_c \simeq 60\text{--}800$ for $z_e = 20$, 230–3100 times the dark matter, where the deposit would exceed the matter density 47-fold at z_e —far past the cap $|\Omega_{\phi,0}| < \Omega_m/(1+z_e) \simeq 0.015$ that $H^2 > 0$ imposes. Within the dark-matter budget the deposit reaches at most 3×10^{-3} ; for one seed per massive galaxy it is of the order of Ω_{GW} , and at z_e a fraction $|\rho_\phi|/\rho_m = n|E_\phi|/\rho_{m,0} \simeq 5 \times 10^{-13}\text{--}6 \times 10^{-8}$ of the matter density, so it does not back-react on the expansion.

Nothing in the astrophysical mass range survives (Sec. IV F), so every observed supermassive black hole is at most a descendant of this channel, and fits of a static Ellis–Bronnikov metric to the Sgr A* or M87* shadows are excluded within this model. The claim is bounded: collapse, alone or in a merger, is a heavy-seed channel with one measured property—prompt conversion at comparable mass—two that follow from a primordial birth—halo independence and no μ -distortion—and one testable signature: the encounter bursts, which LISA would resolve one by one. Whether it is *the* channel behind the JWST population is a question of abundance, and the abundance is cheap: one conversion per little red dot, $n \simeq (3\text{--}6) \times 10^{-5} \text{Mpc}^{-3}$ (the number densities of Refs. [48, 49], summed over their bins), is a mass density $nM \simeq 0.3\text{--}60 M_\odot \text{Mpc}^{-3}$ at $10^4\text{--}10^6 M_\odot$, a fraction $10^{-11}\text{--}2 \times 10^{-9}$ of the dark matter (Fig. 9(c)). LISA would then record 0.06–0.1 bursts in four years, so it tests the channel only if conversions outnumber the little red dots at least tenfold. The evolutions map a birth abundance $n(M)$ onto these observables; they do not fix it (Sec. XD).

C. The inflating half of the population

Seeding is half the census. The branch is chosen by the perturbation’s sign (Sec. IV C), so a sign-symmetric primordial seed distribution converts about half of a foam population into black holes and sets the other half inflating; only large seeds skew the count, and toward collapse—at $\varepsilon = \pm 0.1$ both signs collapse (Sec. IV C). What an inflating throat grows is a baby universe in the

sense of Refs. [71, 72]: the anti-trapped region behind its $\theta_k = 0$ boundary inflates (Sec. IV E) while the exterior remains asymptotically flat at the ADM mass of its initial data, so the parent universe does not expand—Roman’s inflating-wormhole picture [4], reached nonlinearly and without a false-vacuum bubble. The record demonstrates the mechanism, not the magnitude: by $t = 218$ the neck accumulates 1.3 areal e-folds and the boundary 3.0, against the ~ 60 of cosmic inflation [73].

The boundary’s speed turns this loose end into a bound. The $\theta_k = 0$ surface advances at $0.33c$ in areal radius, with no slowing over the record’s last fifty units, and two inflating mouths that meet end as an expanding cloud of the field (Sec. VII A). A population of such mouths must not tile the universe by today. Light has crossed $D_c = 11$ Gpc comoving since $z_e \simeq 20$, so mouths of comoving density $n = 10^{-4}\text{--}10^{-2} \text{Mpc}^{-3}$ —the seed abundances of Sec. XB, which a symmetric census assigns to the inflating half as well—first overlap at mean comoving expansion speeds of $(2.6 \times 10^{-4}\text{--}1.2 \times 10^{-3})c$. The seed channel is therefore consistent only if the boundary stalls after the measured window—the recorded speed is 270 times the loosest bound—or the inflating half is far rarer than the collapsing half. Nothing in the record shows the onset of either; the throat’s end state (Sec. IV E), evolved in a slicing that does not freeze the neck, decides it. In the foam limit the mouths overlap by construction, and a predominantly inflating foam is a candidate pre-inflationary expansion phase [7, 8] with a built-in exit, since growing perturbations widen the collapse basin; the equal-mass, two-throat evolutions here neither model that gas nor fix its abundance.

D. Scope

The results hold for one matter model and one family of data: a single minimally coupled ghost scalar, equal masses, one throat compactness ($m/a = 1/2$), no spin, and superposed initial data whose defect is declared rather than removed (Sec. II C). Horizons here are MOTSs on the evolved slices, which for matter violating the null energy condition need not lie inside event horizons (Sec. III C). The cosmology of Sec. XB takes the birth abundance $n(M)$ and the conversion epoch z_e as inputs—Fig. 9(c) maps any choice onto its signals—and requires the ghost scalar to exist at that epoch.

XI. REPRODUCIBILITY AND COST

Every evolution in Table I is one line in a registry shipped with the results pack—its single knob against a named reference, its outcome, its caveats—and every figure is regenerated by a script in the pack from the archived diagnostic streams. The drainhole initial data, the seeds of Sec. IID, the orientation-corrected surface scan and the run parameter files accompany the reg-

istry; the code is `GRTEclyn` [41]. The evolutions are bit-reproducible on the production hardware: two builds differing only in a passive diagnostic give byte-identical physics streams over 2384 output rows.

Every evolution ran on one NVIDIA H100 80 GB device, with `GRTEclyn` built CUDA-native against AMReX 26.02. At $\Delta t = 0.02 \Delta x$ a production arm ($L = 64$, 128^3 base, three refinement levels) advances 6–18 code units per hour, and a $t = 100$ single-throat or head-on evolution takes 5–7 h. Depth is the expense: level-5 arms run at 2–4 units per hour on the full box (the octant $L = 512$ arm, an eighth of the cells, at 28–74, 11.9 h in all)—28.2 h for the head-on’s $t = 0 \rightarrow 100$ and 46.7 h for the fly-by on the doubled box, which peaks at 58 of the 80 GB—and the production spiral’s three-stage chain costs 23.5 h. The 133 evolutions total 663 GPU-hours, recomputed from each run’s logged speed and time span, a restarted leg from its checkpoint; the shakedown adds 9 h.

ACKNOWLEDGMENTS

This work was supported by Gravity Frontiers (<https://www.gravityfrontiers.org/en>), which funded the underlying research and the development of the numerical-simulation software. The authors acknowledge the GRTL Collaboration for `GRChombo`, `GRtresna`, and `GRTEclyn`.

Appendix A: Code Health and Constraint Evolution

This appendix collects the checks that the evolutions are numerically sound, so that the main-text figures show physics only. Figure 10 gives the constraint record of every run drawn in Figs. 1, 2, 3, 5 and 6, each at the highest refinement level it was run at. Figure 11 shows the horizon regrowth in the single-throat scans that Sec. IV D identifies as numerical; Fig. 12, the spiral’s curvature wall under refinement, gauge and start time (Sec. VII C); and Fig. 13, the spiral burst’s independence of the interior fill (Secs. VII B and VIII C).

Appendix B: Supplementary Measurements

Five measurements support the main text without being needed to follow it: the momentum scan that separates merger from fly-by (Fig. 14; Sec. VII A), the growth of the merging and scattering binaries’ mouths before contact (Fig. 15; Sec. VII B), the linearity of the collapsing throat’s wave in its quadrupole seed (Fig. 16; Sec. VIII A), the scalar channel’s energy, multipoles and validity window on the Ψ_4 spheres (Fig. 17; Sec. VIII F), and the horizon’s shut-off of that channel (Fig. 18; Sec. VIII G).

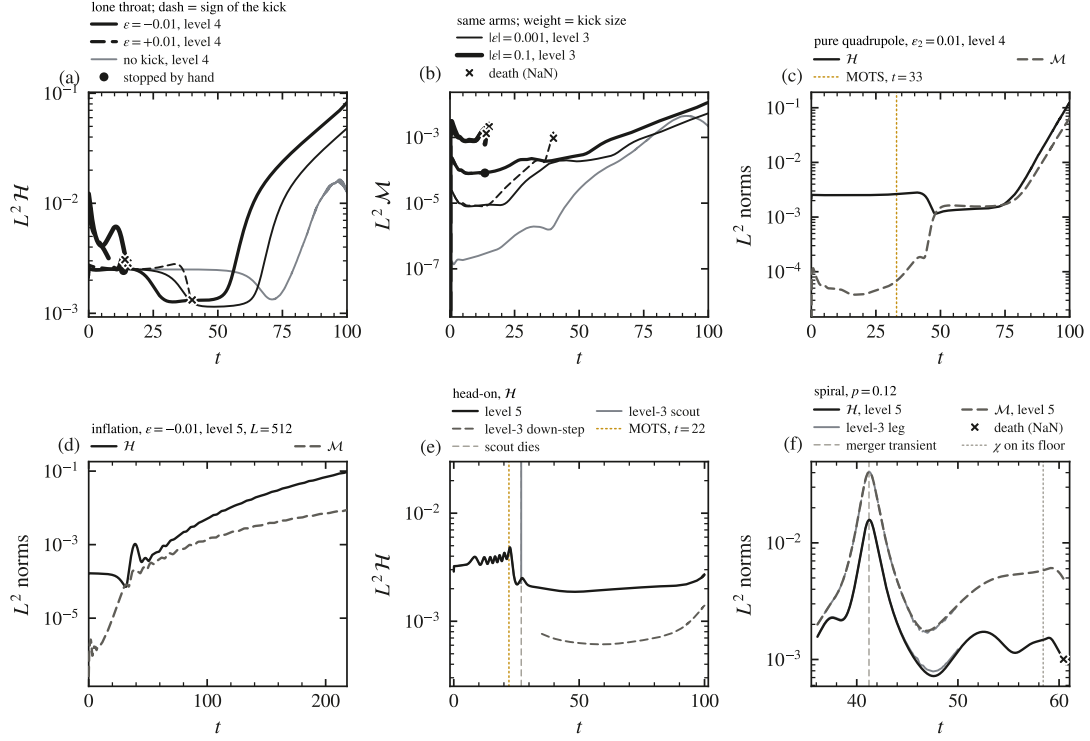


FIG. 10. Constraint evolution of every run the main-text figures draw, each at the highest refinement level it was run at (named in each panel's key): L^2 norms of the Hamiltonian ($\mathcal{H}$, ink) and momentum ($\mathcal{M}$, grey dashed) constraints. The norms are box averages on the base grid ($\Delta x = 0.5$) and compare only within one box (Sec. III C). (a),(b) The lone throat of Fig. 1: level 4 for the unkicked throat and $\varepsilon = \pm 10^{-2}$ ($+10^{-2}$ stopped by hand at $t = 13.5$, dot), level 3 for $\pm 10^{-3}$ and ± 0.1 , the only level these were run at (dash = sign, weight = amplitude, grey = no kick, crosses = deaths). The base grid does not resolve the seed's shell: at $t = 0$ the norms carry the exact data's discretisation floor, to which a kick adds its own defect in quadrature, $0.090 |\varepsilon|$ ($2.1\text{--}2.35 \times 10^{-3}$ for $|\varepsilon| \leq 10^{-2}$), although on the shell it reaches $0.93 \times 16\pi |\rho|$ at 10^{-2} (Sec. II D). $\mathcal{M}$ vanishes at $t = 0$ and is fed by that defect within the first steps, in proportion to ε (2.3×10^{-4} at 10^{-2} , 2.3×10^{-5} at 10^{-3} , at $t = 0.5$, against 2×10^{-7} unkicked). At level 3 the $|\varepsilon| \leq 10^{-2}$ arms then follow the unkicked norms to 5.4% until the fates part—a bound on the box average, not on the violation near the throat (Sec. IV D); $\varepsilon = \pm 0.1$ starts $3\text{--}6\times$ high and dies at $t = 14.1$ ($+0.1$) and 15.2 (-0.1) in a one-step overflow at the compactified origin. (c) The pure-quadrupole collapse of Fig. 2 (level 4; gold, its MOTS from $t = 33$). The rise from $t \gtrsim 75$ is a grid mode, neither this collapse nor a reflection off the box wall: it starts on the refinement boundaries, every single-throat arm that runs long enough shows it, collapsing or inflating, and the doubled box does not delay it: the norms leave their floor from $t \approx 77$ ($L = 128$) and 82 ($L = 64$). (d) The inflating throat of Fig. 3 (level 5, $L = 512$) to $t = 218$. (e) The head-on of Fig. 5, $\mathcal{H}$ only: the level-5 arm stays below its value at formation (gold, $t = 22$) to $t = 100$, while the level-3 scout's norm leaves the frame at its wall ($t = 26.91$, grey dashed); dashed, the level-3 down-step. (f) The production spiral of Fig. 6: the level-5 leg to its NaN at $t = 60.45$ (cross), and the level-3 leg over their common window, which it reproduces to a median 3.5%/0.7%; rules, the merger transient ($t = 41.2$) and χ on its floor ($t = 58.4$). At the NaN both norms sit below their transient peaks: the wall is not a constraint blow-up.

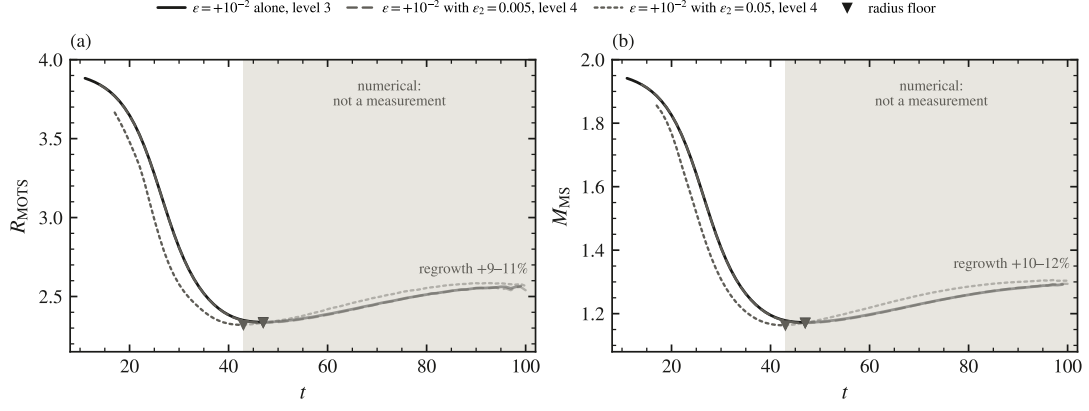


FIG. 11. The remnant horizon of the $\varepsilon = +10^{-2}$ collapse: MOTS areal radius (a) and Misner–Sharp mass (b) on the throat-centred scan. All three arms carry the radial kick $\varepsilon = +10^{-2}$: alone at level 3 and, at level 4, dressed with a quadrupole $\varepsilon_2 = 0.005$ or 0.05 (floors marked). The fall, by 40% in radius, is the phantom being swallowed. Past the first floor (shaded, curves faint) the scans follow a constraint-violation layer that has reached the MOTS, not the horizon: the rise there, 9–11% in radius and 10–12% in mass, is numerical and carries no result, since in spherical symmetry a phantom can only shrink a horizon (Sec. IV D). The pure quadrupole, which carries no radial kick, only shrinks (Fig. 2(a)).

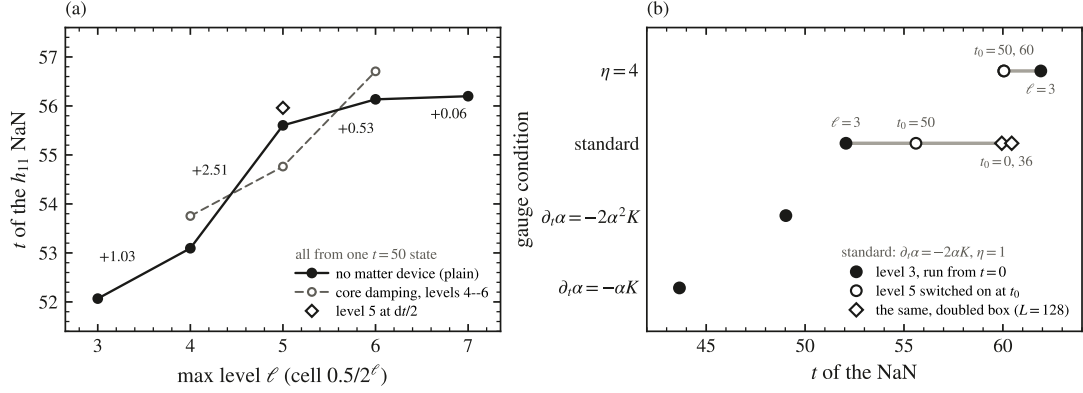


FIG. 12. Death time of the $p = 0.12$ spiral under refinement, gauge and start time; every death is an h_{11} NaN except in the $\partial_t \alpha = -\alpha K$ arm, which fails in K . (a) Refinement from one $t = 50$ state of the $L = 64$ run, maximum level ℓ (finest cell $0.5/2^\ell$). Filled circles: no matter device, gain per doubling marked; open circles: core matter damping (levels 4–6 share one configuration); open diamond: level 5 at half the time step. (b) The same wall under other gauges and start times, one row per gauge. The lapse obeys $\partial_t \alpha = -c \alpha^p (K - 2\Theta)$ and the shift is damped at rate η ; the standard gauge is $c = 2$, $p = 1$, $\eta = 1$, and each other row changes one of the three (and carries the core matter damping, worth under a unit; panel a). Filled circles: level 3 from $t = 0$; open circles: level 5 from t_0 on $L = 64$; open diamonds: level 5 on $L = 128$. Level 3: $t = 43.6, 49.0, 52.1, 61.9$. Level 5: standard gauge 59.94 and 60.45 on $L = 128$ ($t_0 = 0, 36$) and 55.60 on $L = 64$ ($t_0 = 50$); $\eta = 4$, 60.04 and 60.05 ($t_0 = 50, 60$).

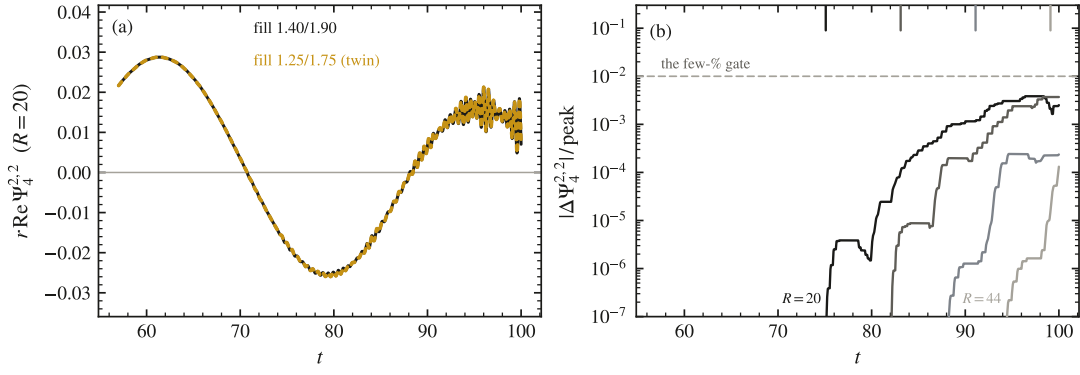


FIG. 13. Insensitivity of the spiral’s burst to the fill window. (a) $r \operatorname{Re} \Psi_4^{2,2}$ at $R = 20$ for the freeze arm (fill 1.40/1.90, ink) and its twin (1.25/1.75, gold dots) from the same $t = 57$ checkpoint. (b) Per-sphere envelope of the difference: at most 0.385% of peak at $R = 20$ and 0.013% at $R = 44$, with ticks at the fill’s causal clock $t = 57 + (R - 1.9)$ on each sphere. The difference reaches $R = 20$ on that clock and runs ahead of it farther out at $1.2\text{--}1.3c$ (a gauge-speed front), below 1.5×10^{-5} of peak. The outcome: moving the frozen region’s edge moves the waveform by under half a percent, and the burst, which peaks at $R = 20$ 13 units before the core’s light cone arrives, cannot depend on the freeze at all.

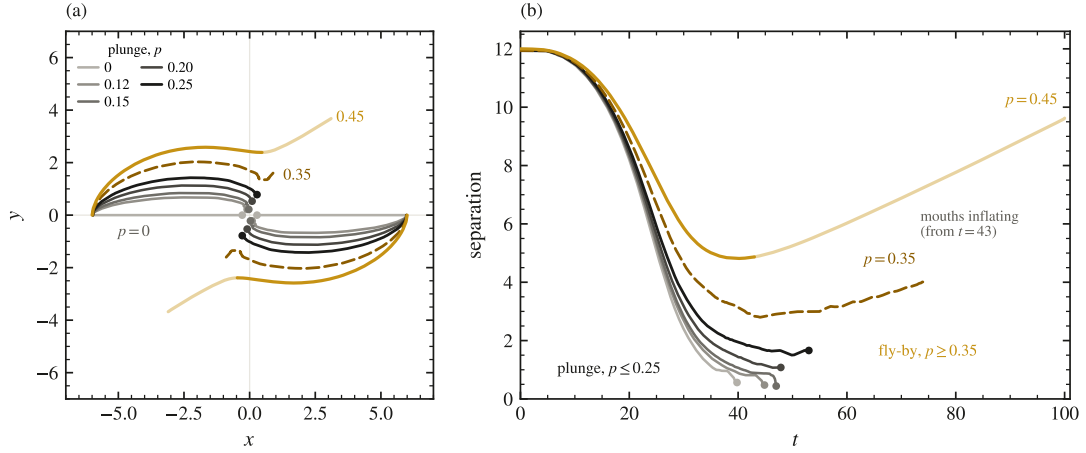


FIG. 14. Merger or fly-by: the momentum scan from $d = 12$ with opposite signs, from the head-on release ($p = 0$) to the fly-by ($p = 0.45$). (a) Both throats in the orbital plane; (b) their separation. The tracks are the coordinate positions of the χ pits, followed through each run. Grey, light to dark: the plunges $p = 0, 0.12, 0.15, 0.20$ and 0.25 , each drawn until χ first reaches its floor or the run ends (dots); gold: the fly-bys $p = 0.35$ (dashed) and 0.45 . Every arm is level 3 except the $p = 0.45$ fly-by, the level-5 arm of the text; since a level-3 pit sits on a cell of side $1/16$, the tracks are averaged over two time units. The fly-by turns at closest approach, 4.8 at $t \approx 40$; from $t \approx 43$ its mouths have inflated past the scan window, and the faint stretch is the separation of the χ pits of two inflating mouths, not of two compact throats.

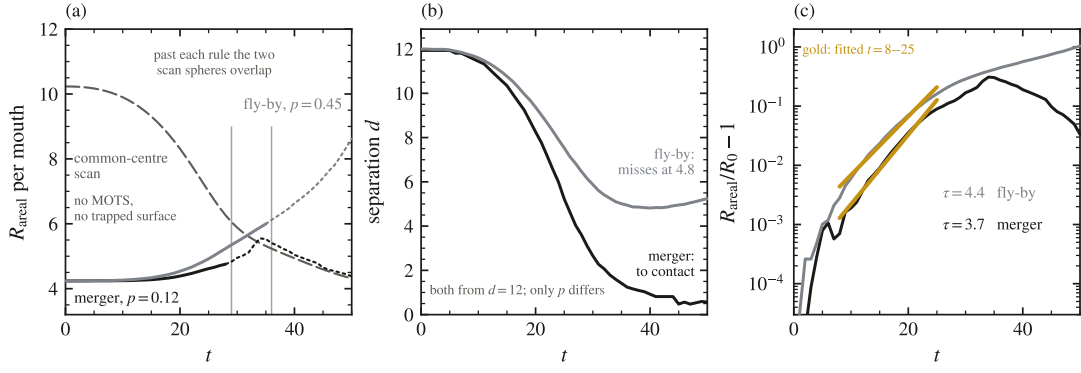


FIG. 15. Mouth growth in the merging ($p = 0.12$) and scattering ($p = 0.45$) arms, both from $d = 12$ with the same throats; the merger's instrumented re-run carries three refinement levels and the fly-by five, so the shared τ is a result across two resolutions rather than a one-knob comparison. (a) Areal radius of one mouth (its twin agrees to 2×10^{-4}): solid while the two scan spheres are disjoint, dotted once they overlap; long dash, the common-centre scan. No trapped surface appears on these star-shaped scans, which end at $t = 50$ (merger) and $t = 100$ (fly-by). (b) Separation: $p = 0.12$ reaches contact, $p = 0.45$ misses at 4.8. (c) Growth excess $R/R_0 - 1$ with exponential fits over $t = 8-25$ (gold): $\tau = 3.70$ (merger) and 4.39 (fly-by).

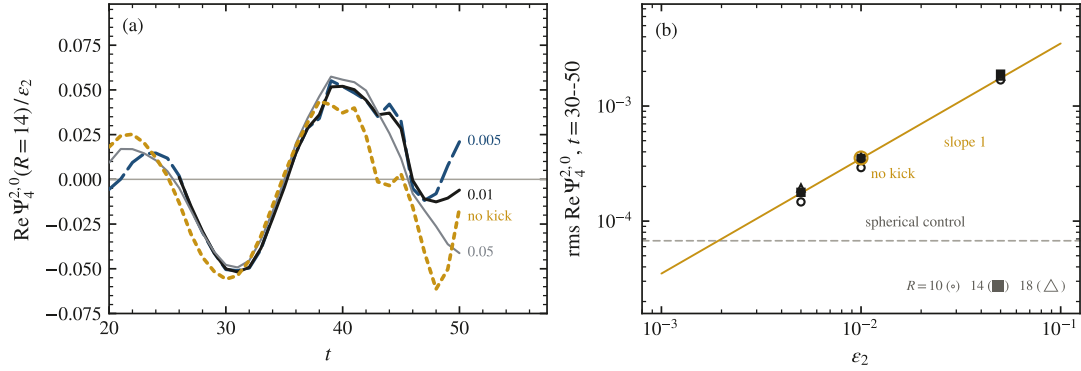


FIG. 16. Linearity of the collapsing throat's wave in its quadrupole seed. (a) $\text{Re } \Psi_4^{2,0}$ at $R = 14$, each arm divided by its seed and named at its end: $\epsilon_2 = 0.005, 0.01, 0.05$ and the kick-free 0.01 arm collapse onto one curve through the burst; later the smallest arm carries its floor amplified by $1/\epsilon_2$ and the kick-free arm runs ahead. (b) rms amplitude over $t = 30-50$ against ϵ_2 at three spheres, with the slope-one line through the 0.01 arm drawn across the decade $10^{-3}-10^{-1}$ as a reference. Open circle: the pure quadrupole, within 2% of the kicked arm. Dashed rule: the floor of the level-4 spherical control (the same kick, no quadrupole) at $R = 14$, which the line crosses at $\epsilon_2 \approx 2 \times 10^{-3}$. The outcome: the wave is the quadrupole's—linear in ϵ_2 and unchanged, to 2% in amplitude, when the radial kick is removed. The kick alone radiates only the floor; it moves the wave's phase, not its size.

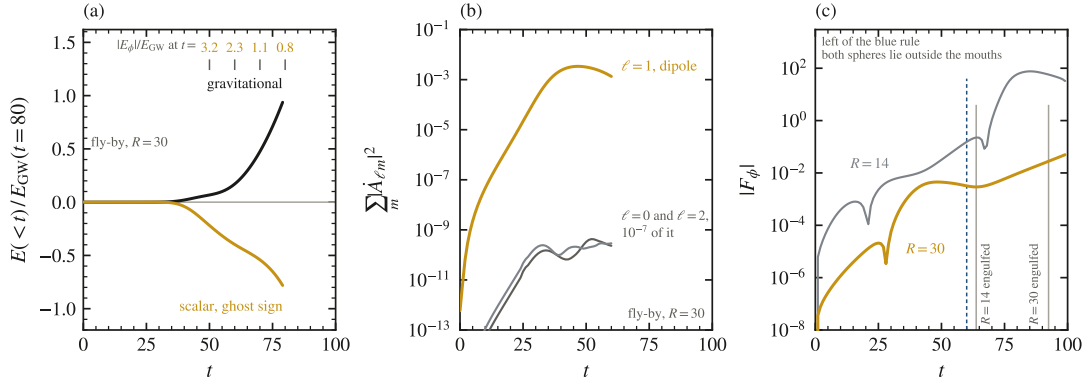


FIG. 17. The scalar channel on the Ψ_4 spheres. (a) Fly-by energy through $R = 30$ in units of the gravitational energy crossed by $t = 80$: gravitational (ink) and scalar with its physical sign (gold, Sec. VIII F). The ratio $|E_\phi|/E_{\text{GW}}$ printed at each cut falls from 3.2 to 0.8 as the gravitational burst arrives and this running integral drifts (Sec. VIII F); past $t \approx 92$ the mouths' areal-radius reading, an upper bound, passes the sphere. (b) Multipole content, fly-by: $\ell = 1$ exceeds $\ell = 0$ and $\ell = 2$ by 10^7 . (c) Validity window: $|F_\phi|$ at both spheres, with the times at which the mouths' areal-radius reading (an upper bound) passes each sphere (grey rules); left of the blue rule both spheres lie outside the mouths. The $t = 60$ window at $R = 14$ gives 3.0 in place of 2.3.

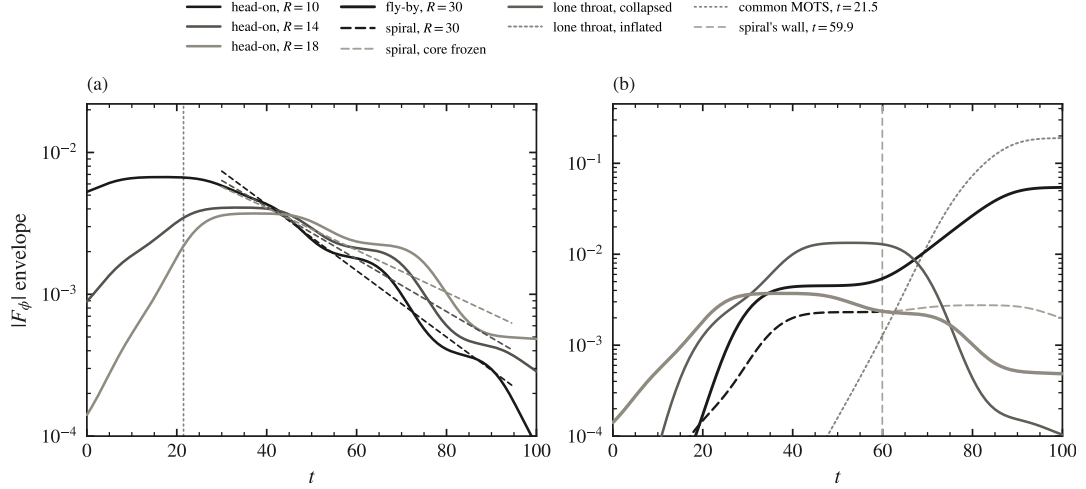


FIG. 18. Envelope of the scalar flux $|F_\phi|$, horizonized against horizonless arms: a running maximum over 25 units (one dipole period), smoothed in log amplitude, since the raw flux changes sign every ~ 20 units. (a) The head-on (level 5 from $t = 0$) at $R = 10/14/18$. The common MOTS closes at $t = 21.5$ (dotted); the remaining hair leaves as one $\ell = 1$ pulse, cresting later at each sphere, then decays (dashed: fits over $t = 30-95$, $\tau = 19/24/29$). The rise before the rule is the open mouths' static hair, canonically ingoing; only the post-horizon window is integrated. (b) Each encounter on its outermost sphere (head-on and lone throat $R = 18$, fly-by and spiral $R = 30$): the fly-by grows to the end of its record; the spiral, with a horizon from $t = 55.0$ and too young to shed on a ~ 20 clock, is still climbing at its wall ($t = 59.9$, dashed rule), beyond which the frozen-core arm is drawn dashed and carries no claim. Mid-grey: the lone throat twice—the $\varepsilon = +10^{-2}$ level-4 collapse (solid), behind a MOTS from $t = 11$, and the unkicked level-4 throat (dotted), which inflates, horizonless, and grows in the monopole (counted to $t \approx 80$).

-
- [1] J. A. Wheeler, “Geons,” *Phys. Rev.* **97**, 511 (1955).
 - [2] J. A. Wheeler, “On the nature of quantum geometrodynamics,” *Ann. Phys. (N.Y.)* **2**, 604 (1957).
 - [3] C. W. Misner and J. A. Wheeler, “Classical physics as geometry,” *Ann. Phys. (N.Y.)* **2**, 525 (1957).
 - [4] T. A. Roman, “Inflating Lorentzian wormholes,” *Phys. Rev. D* **47**, 1370 (1993).
 - [5] J. Garriga, A. Vilenkin, and J. Zhang, “Black holes and the multiverse,” *J. Cosmol. Astropart. Phys.* **02**, 064 (2016).
 - [6] H. Deng, J. Garriga, and A. Vilenkin, “Primordial black hole and wormhole formation by domain walls,” *J. Cosmol. Astropart. Phys.* **04**, 050 (2017).
 - [7] A. A. Kirillov and E. P. Savelova, “Dark matter from a gas of wormholes,” *Phys. Lett. B* **660**, 93 (2008).
 - [8] A. A. Kirillov and E. P. Savelova, “Density perturbations in a gas of wormholes,” *Mon. Not. R. Astron. Soc.* **412**, 1710 (2011).
 - [9] H. G. Ellis, “Ether flow through a drainhole: A particle model in general relativity,” *J. Math. Phys.* **14**, 104 (1973).
 - [10] K. A. Bronnikov, “Scalar-tensor theory and scalar charge,” *Acta Phys. Polon. B* **4**, 251 (1973).
 - [11] M. S. Morris and K. S. Thorne, “Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity,” *Am. J. Phys.* **56**, 395 (1988).
 - [12] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (AIP Press, Woodbury, NY, 1995).
 - [13] H. Shinkai and S. A. Hayward, “Fate of the first traversable wormhole: Black-hole collapse or inflationary expansion,” *Phys. Rev. D* **66**, 044005 (2002).
 - [14] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. II. Nonlinear evolution,” *Class. Quantum Grav.* **26**, 015011 (2009).
 - [15] J. A. González, F. S. Guzmán, and O. Sarbach, “Instability of wormholes supported by a ghost scalar field. I. Linear stability analysis,” *Class. Quantum Grav.* **26**, 015010 (2009).
 - [16] R. Geroch, “Topology in general relativity,” *J. Math. Phys.* **8**, 782 (1967).
 - [17] F. J. Tipler, “Singularities and causality violation,” *Ann. Phys. (N.Y.)* **108**, 1 (1977).
 - [18] S. W. Hawking, “Wormholes in spacetime,” *Phys. Rev. D* **37**, 904 (1988).
 - [19] Ya. B. Zel’dovich and I. D. Novikov, “The hypothesis of cores retarded during expansion and the hot cosmological model,” *Sov. Astron.* **10**, 602 (1967).
 - [20] B. J. Carr and S. W. Hawking, “Black holes in the early Universe,” *Mon. Not. R. Astron. Soc.* **168**, 399 (1974).
 - [21] B. Carr, K. Kohri, Y. Sendouda, and J. Yokoyama, “Constraints on primordial black holes,” *Rep. Prog. Phys.* **84**, 116902 (2021).
 - [22] C. Caprini and D. G. Figueroa, “Cosmological backgrounds of gravitational waves,” *Class. Quantum Grav.* **35**, 163001 (2018).
 - [23] G. Agazie *et al.* (NANOGrav Collaboration), “The NANOGrav 15 yr data set: Evidence for a gravitational-wave background,” *Astrophys. J. Lett.* **951**, L8 (2023).
 - [24] P. Amaro-Seoane *et al.*, “Laser Interferometer Space Antenna,” arXiv:1702.00786 [astro-ph.IM] (2017).
 - [25] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Upper limits on the isotropic gravitational-wave background from Advanced LIGO and Advanced Virgo’s third observing run,” *Phys. Rev. D* **104**, 022004 (2021).
 - [26] L. Pagano, L. Salvati, and A. Melchiorri, “New constraints on primordial gravitational waves from Planck 2015,” *Phys. Lett. B* **760**, 823 (2016).
 - [27] B. Allen, W. G. Anderson, P. R. Brady, D. A. Brown, and J. D. E. Creighton, “FINDCHIRP: An algorithm for detection of gravitational waves from inspiraling compact binaries,” *Phys. Rev. D* **85**, 122006 (2012).
 - [28] C. Reisswig and D. Pollney, “Notes on the integration of numerical relativity waveforms,” *Class. Quantum Grav.* **28**, 195015 (2011).
 - [29] S. A. Usman *et al.*, “The PyCBC search for gravitational waves from compact binary coalescence,” *Class. Quantum Grav.* **33**, 215004 (2016).
 - [30] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “Open data from the third observing run of LIGO, Virgo, KAGRA and GEO,” *Astrophys. J. Suppl.* **267**, 29 (2023).
 - [31] R. Abbott *et al.* (LIGO Scientific, Virgo and KAGRA Collaborations), “GWTC-3: Compact binary coalescences observed by LIGO and Virgo during the second part of the third observing run,” *Phys. Rev. X* **13**, 041039 (2023).
 - [32] C. W. Misner and D. H. Sharp, “Relativistic equations for adiabatic, spherically symmetric gravitational collapse,” *Phys. Rev.* **136**, B571 (1964).
 - [33] D. Hochberg and M. Visser, “Dynamic wormholes, anti-trapped surfaces, and energy conditions,” *Phys. Rev. D* **58**, 044021 (1998).
 - [34] S. W. Hawking, “Gravitational radiation from colliding black holes,” *Phys. Rev. Lett.* **26**, 1344 (1971).
 - [35] J. M. Bowen and J. W. York, “Time-asymmetric initial data for black holes and black-hole collisions,” *Phys. Rev. D* **21**, 2047 (1980).
 - [36] T. Helfer, U. Sperhake, R. Croft, M. Radia, B.-X. Ge, and E. A. Lim, “Malaise and remedy of binary boson-star initial data,” *Class. Quantum Grav.* **39**, 074001 (2022).
 - [37] D. Alic, C. Bona-Casas, C. Bona, L. Rezzolla, and C. Palenzuela, “Conformal and covariant formulation of the Z4 system with constraint-violating solutions,” *Phys. Rev. D* **85**, 064040 (2012).
 - [38] C. Bona, J. Massó, E. Seidel, and J. Stela, “New formalism for numerical relativity,” *Phys. Rev. Lett.* **75**, 600 (1995).
 - [39] M. Campanelli, C. O. Lousto, P. Marronetti, and Y. Zlochower, “Accurate evolutions of orbiting black-hole binaries without excision,” *Phys. Rev. Lett.* **96**, 111101 (2006).
 - [40] J. G. Baker, J. Centrella, D.-I. Choi, M. Koppitz, and J. van Meter, “Gravitational-wave extraction from an inspiraling configuration of merging black holes,” *Phys. Rev. Lett.* **96**, 111102 (2006).
 - [41] GRTL Collaboration, *GRTEclyn*, <https://github.com/GRTLCollaboration/GRTEclyn>.
 - [42] W. Zhang *et al.*, “AMReX: a framework for block-structured adaptive mesh refinement,” *J. Open Source Softw.* **4**, 1370 (2019).

- [43] T. Andrade *et al.*, “GRChombo: an adaptable numerical relativity code for fundamental physics,” *J. Open Source Softw.* **6**, 3703 (2021).
- [44] N. M. Shirokov, “Wormhole dynamics: nonlinear collapse and gravitational-wave emission,” arXiv:2604.00071 [gr-qc] (2026).
- [45] N. M. Shirokov, “The Bondi dipole in full numerical relativity: a self-accelerating positive–negative mass binary,” arXiv:2608.24577 [gr-qc] (2026).
- [46] Á. Bogdán *et al.*, “Evidence for heavy-seed origin of early supermassive black holes from a $z \approx 10$ X-ray quasar,” *Nat. Astron.* **8**, 126 (2024).
- [47] P. Natarajan, F. Pacucci, A. Ricarte, Á. Bogdán, A. D. Goulding, and N. Cappelluti, “First detection of an overmassive black hole galaxy: UHZ1—evidence for heavy black hole seeds from direct collapse?,” *Astrophys. J. Lett.* **960**, L1 (2024).
- [48] J. E. Greene *et al.*, “UNCOVER spectroscopy confirms the surprising ubiquity of active galactic nuclei in red sources at $z > 5$,” *Astrophys. J.* **964**, 39 (2024).
- [49] J. Matthee *et al.*, “Little red dots: an abundant population of faint active galactic nuclei at $z \sim 5$ revealed by the EIGER and FRESCO JWST surveys,” *Astrophys. J.* **963**, 129 (2024).
- [50] R. Maiolino *et al.*, “A small and vigorous black hole in the early Universe,” *Nature* **627**, 59 (2024).
- [51] E. Bañados *et al.*, “An 800-million-solar-mass black hole in a significantly neutral Universe at a redshift of 7.5,” *Nature* **553**, 473 (2018).
- [52] K. Inayoshi, E. Visbal, and Z. Haiman, “The assembly of the first massive black holes,” *Annu. Rev. Astron. Astrophys.* **58**, 27 (2020).
- [53] T. Nakama, B. Carr, and J. Silk, “Limits on primordial black holes from μ distortions in cosmic microwave background,” *Phys. Rev. D* **97**, 043525 (2018).
- [54] M. Alcubierre, “Hyperbolic slicings of spacetime: singularity avoidance and gauge shocks,” *Class. Quantum Grav.* **20**, 607 (2003).
- [55] K. Clough, “Continuity equations for general matter: applications in numerical relativity,” *Class. Quantum Grav.* **38**, 167001 (2021).
- [56] E. Gourgoulhon, *3+1 Formalism in General Relativity: Bases of Numerical Relativity*, Lecture Notes in Physics Vol. 846 (Springer, Berlin, 2012).
- [57] B. Kleihaus and J. Kunz, “Rotating Ellis wormholes in four dimensions,” *Phys. Rev. D* **90**, 121503(R) (2014).
- [58] V. Dzhunushaliev, V. Folomeev, B. Kleihaus, J. Kunz, and E. Radu, “Rotating wormholes in five dimensions,” *Phys. Rev. D* **88**, 124028 (2013).
- [59] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, “Are slowly rotating Ellis-Bronnikov wormholes stable?,” *Phys. Lett. B* **848**, 138349 (2024).
- [60] B. Azad, J. L. Blázquez-Salcedo, F. S. Khoo, and J. Kunz, “Radial perturbations of Ellis-Bronnikov wormholes in slow rotation up to second order,” *Phys. Rev. D* **109**, 124051 (2024).
- [61] F. S. Khoo, B. Azad, J. L. Blázquez-Salcedo, L. M. González-Romero, B. Kleihaus, J. Kunz, and F. Navarro-Lérida, “Quasinormal modes of rapidly rotating Ellis-Bronnikov wormholes,” *Phys. Rev. D* **109**, 084013 (2024).
- [62] E. Poisson and M. Visser, “Thin-shell wormholes: Linearization stability,” *Phys. Rev. D* **52**, 7318 (1995).
- [63] P. Kanti, B. Kleihaus, and J. Kunz, “Wormholes in dilatonic Einstein-Gauss-Bonnet theory,” *Phys. Rev. Lett.* **107**, 271101 (2011).
- [64] K. A. Bronnikov, J. C. Fabris, and A. Zhidenko, “On the stability of scalar-vacuum space-times,” *Eur. Phys. J. C* **71**, 1791 (2011).
- [65] S. A. Hayward, “Gravitational energy in spherical symmetry,” *Phys. Rev. D* **53**, 1938 (1996).
- [66] E. Babichev, V. Dokuchaev, and Yu. Eroshenko, “Black hole mass decreasing due to phantom energy accretion,” *Phys. Rev. Lett.* **93**, 021102 (2004).
- [67] T. Robson, N. J. Cornish, and C. Liu, “The construction and use of LISA sensitivity curves,” *Class. Quantum Grav.* **36**, 105011 (2019).
- [68] E. Thrane and J. D. Romano, “Sensitivity curves for searches for gravitational-wave backgrounds,” *Phys. Rev. D* **88**, 124032 (2013).
- [69] N. Aghanim *et al.* (Planck Collaboration), “Planck 2018 results. VI. Cosmological parameters,” *Astron. Astrophys.* **641**, A6 (2020).
- [70] A. Ashtekar and B. Krishnan, “Isolated and dynamical horizons and their applications,” *Living Rev. Relativ.* **7**, 10 (2004).
- [71] E. Farhi and A. H. Guth, “An obstacle to creating a universe in the laboratory,” *Phys. Lett. B* **183**, 149 (1987).
- [72] S. K. Blau, E. I. Guendelman, and A. H. Guth, “Dynamics of false-vacuum bubbles,” *Phys. Rev. D* **35**, 1747 (1987).
- [73] A. R. Liddle and S. M. Leach, “How long before the end of inflation were observable perturbations produced?,” *Phys. Rev. D* **68**, 103503 (2003).